

ZTE ZXA10 C300 INSTALLATION MANUAL



Quick Links

[Cabinet Introduction](#)

[System Components](#)

Table of Contents

- Table of Contents
- About This Manual
- Cabinet Introduction
- Cabinet Features
 - Figure 1-1 Cabinet Appearance
 - Figure 1-2 Cabinet Appearance
- Equipment Configuration
 - Cabinet Configuration
 - Shelf Configuration
 - Figure 1-3 Cabinet Configuration for 19-Inch Shelves
 - Figure 1-4 19-Inch Shelf Structure
 - Figure 1-5 19-Inch Shelf Structure
 - Cards
 - Figure 1-6 19-Inch Shelf Configuration
 - Table 1-1 19-Inch Shelf Technical Specifications
 - Table 1-2 Card List
- System Components
- Technical Standards
 - Table 1-3 Technical Standards
- Technical Specifications
 - Performance Specifications
 - Table 1-4 Reliability and Security Specifications
- Installation Preparation Flow
 - Engineering Documents Preparation
 - Figure 2-1 Installation Preparation Flow
- Tools and Meters
- Installation Environment Check
 - Construction Check
 - Indoor Environment Check
 - Table 2-1 Operation Environment Requirements
 - Power Supply Check
 - Grounding Cable Check
 - Security Check

Other Supporting Facilities Check

Unpacking

Unpacking Cabinet

Figure 2-2 Packing Box

Figure 2-3 Cabinet Inner Package

Unpacking Shelf

Figure 2-4 Shelf Packing Box

Unpacking Cards

Figure 2-5 Card Packing Box

Figure 2-6 Opened Card Packing Box

Cabinet Installation Flowchart

Base Installation Mode

Base Structure

Figure 3-1 Cabinet Installation Flowchart

Figure 3-2 Cabinet Base Structure

Figure 3-3 Base With Adjustable Height From 160 mm to 260 mm

Figure 3-4 Base With Adjustable Height From 260 mm to 460 mm

Figure 3-5 Base With Adjustable Height From 460 mm to 660 mm

Base Installation Flowchart

Positioning Base

Figure 3-6 Base Installation Flowchart

Figure 3-7 Drilling Diagram

Installing Base

Fixing Cabinet

Figure 3-8 Installing Base

Figure 3-9 Fixing Cabinet

Foot Installation Mode

Foot Installation Flowchart

Figure 3-10 Cabinet Installed on the Base

Positioning Cabinet

Figure 3-11 Foot Installation Flowchart

Figure 3-12 Drilling Diagram

Installing Expansion Bolt

Fixing Cabinet

Figure 3-13 Cabinet Installation Holes

Cabinets Connection

Cabinets Connection in Base Installation Mode

Figure 3-14 Fixing Cabinet

Figure 3-15 Fixing Two Cabinets In Parallel

Cabinets Connection in Foot Installation Mode

Figure 3-16 Fixing Multiple Cabinets In Parallel

Figure 3-17 Fixing Two Cabinets In Parallel

Adjacent Cabinets Connection

Figure 3-18 Fixing Multiple Cabinets In Parallel

Shelf Installation

Figure 3-19 Connecting Adjacent Cabinets

Plug-in Box Installation

Figure 3-20 Installing Shelf

Card Installation

Figure 3-21 Installing Plug-in Box

Power and Grounding Cables

Power and Grounding Cable Structure

Figure 4-1 Power and Grounding Cables Installation Flowchart

Figure 4-2 -48 V Power Cable Structure

Figure 4-3 Internal Power Cable Structure

Table 4-1 Internal Power Cable Pin Connections

Power and Grounding Cable Installation

19-Inch Power Distributor

Figure 4-4 Protection Grounding Cable Structure

Figure 4-5 19-Inch Power Distributor Dimensions

48 V Input Power Cable Installation

Figure 4-6 Power Distributor Internal Structure

Internal Power Cable Installation

Figure 4-7 Power Distributor Cable Connections

Grounding Cable Installation

Figure 4-8 Internal Power Cable Layout

Figure 4-9 Cabinet Grounding Point

Figure 4-10 Protection Ground Connection

Overall Cabling

Cable Layout

Cabling Principles

Figure 5-1 Uplink Cabling
Cable Bundling

Figure 5-2 Cables Bundling

Figure 5-3 Tie Cutting

Figure 5-4 Tie Space

Figure 5-5 Cable Tie Position

E1/T1 Cable

Figure 5-6 Cabling Technique In Cabinet

Figure 5-7 E1 Cable Structure

Table 5-1 E1 Cable Pin Connections

Figure 5-8 T1 Cable Structure

Table 5-2 T1 Cable Pin Connections

Monitoring Cables

Entrance Control Monitoring Cable

Figure 5-9 Monitoring Cable Routing

Smog Monitoring Cable

Figure 5-10 Entrance Control Monitoring Cable Structure

Table 5-3 Entrance Control Monitoring Cable Pin Connections

Flood Monitoring Cable

Figure 5-11 Smog Monitoring Cable Structure

Table 5-4 Smog Monitoring Cable Pin Connections

Temperature Monitoring Cable

Figure 5-12 Flood Monitoring Cable Structure

Table 5-5 Flood Monitoring Cable Pin Connections

Humidity Monitoring Cable

Figure 5-13 Temperature Monitoring Cable Structure

Table 5-6 Temperature Monitoring Cable Pin Connections

Serial Port Cable

Figure 5-14 Humidity Monitoring Cable Structure

Table 5-7 Humidity Monitoring Cable Pin Connections

Figure 5-15 Serial Port Cable Structure

Table 5-8 Serial Port Cable Connections

Network Cable

Figure 5-16 Network Cable Structure

Optical Fiber

Table 5-9 Relationship Between Network Cables and Core Cables

Overall Inspection

Power-ON/OFF Inspection

Inspection before Power-ON

Power-ON Process

Power-OFF Process

Overview

Central Office Cabinet Grounding

Remote Office Cabinet Grounding

Figure A-3 Remote Office Cabinet Grounding

Table B-1 Length and Cross-sectional Area Requirements

Cabinet and Central Office

Outgoing/Incoming Cables from/to Central Office

Lightning Requirements

Lightning Design

Table C-1 Lightning Protection Level

Table C-2 Lightning Protection Device Parameters

Lightning Protection Grounding in Central Office

Lightning Protection Grounding in Remote Office

Figures

Glossary

Other ManualsLib Projects



ZXA10 C300

Optical Access Convergence Equipment

Cabinet Installation Guide (19 Inch)

Version: V1.2.1

ZTE CORPORATION
NO. 55, Hi-tech Road South, ShenZhen, P.R.China
Postcode: 518057
Tel: +86-755-26771900
Fax: +86-755-26770801
URL: <http://ensupport.zte.com.cn>
E-mail: support@zte.com.cn

LEGAL INFORMATION

Copyright © 2011 ZTE CORPORATION.

The contents of this document are protected by copyright laws and international treaties. Any reproduction or distribution of this document or any portion of this document, in any form by any means, without the prior written consent of ZTE CORPORATION is prohibited. Additionally, the contents of this document are protected by contractual confidentiality obligations.

All company, brand and product names are trade or service marks, or registered trade or service marks, of ZTE CORPORATION or of their respective owners.

This document is provided “as is”, and all express, implied, or statutory warranties, representations or conditions are disclaimed, including without limitation any implied warranty of merchantability, fitness for a particular purpose, title or non-infringement. ZTE CORPORATION and its licensors shall not be liable for damages resulting from the use of or reliance on the information contained herein.

ZTE CORPORATION or its licensors may have current or pending intellectual property rights or applications covering the subject matter of this document. Except as expressly provided in any written license between ZTE CORPORATION and its licensee, the user of this document shall not acquire any license to the subject matter herein.

ZTE CORPORATION reserves the right to upgrade or make technical change to this product without further notice.

Users may visit ZTE technical support website <http://ensupport.zte.com.cn> to inquire related information.

The ultimate right to interpret this product resides in ZTE CORPORATION.

Revision History

Revision No.	Revision Date	Revision Reason
R1.1	2011-12-31	Version update
R1.0	2011-09-30	First edition

Serial Number: SJ-20110909101511-005

Publishing Date: 2011-12-31 (R1.1)

Contents

About This Manual	I
Chapter 1 Overview	1-1
1.1 Cabinet Introduction	1-1
1.2 Cabinet Features.....	1-2
1.3 Equipment Configuration	1-3
1.3.1 Cabinet Configuration	1-3
1.3.2 Shelf Configuration.....	1-4
1.3.3 Cards	1-6
1.4 System Components	1-7
1.5 Technical Standards	1-8
1.6 Technical Specifications.....	1-10
1.7 Performance Specifications.....	1-11
Chapter 2 Pre-Installation Preparation	2-1
2.1 Installation Preparation Flow	2-1
2.2 Engineering Documents Preparation	2-2
2.3 Tools and Meters.....	2-3
2.4 Installation Environment Check	2-4
2.4.1 Construction Check	2-4
2.4.2 Indoor Environment Check.....	2-4
2.4.3 Power Supply Check	2-5
2.4.4 Grounding Cable Check.....	2-6
2.4.5 Security Check.....	2-6
2.4.6 Other Supporting Facilities Check	2-6
2.5 Unpacking	2-7
2.5.1 Unpacking Cabinet.....	2-7
2.5.2 Unpacking Shelf.....	2-10
2.5.3 Unpacking Cards.....	2-11
Chapter 3 Cabinet Installation.....	3-1
3.1 Cabinet Installation Flowchart	3-1
3.2 Base Installation Mode	3-2
3.2.1 Base Structure	3-2
3.2.2 Base Installation Flowchart	3-5
3.2.3 Positioning Base	3-5

3.2.4 Installing Base	3-7
3.2.5 Fixing Cabinet.....	3-7
3.3 Foot Installation Mode	3-9
3.3.1 Foot Installation Flowchart	3-9
3.3.2 Positioning Cabinet	3-10
3.3.3 Installing Expansion Bolt.....	3-12
3.3.4 Fixing Cabinet.....	3-12
3.4 Cabinets Connection	3-14
3.4.1 Cabinets Connection in Base Installation Mode	3-14
3.4.2 Cabinets Connection in Foot Installation Mode	3-16
3.4.3 Adjacent Cabinets Connection	3-18
3.5 Shelf Installation.....	3-19
3.6 Plug-in Box Installation	3-20
3.7 Card Installation	3-21
Chapter 4 Power Supply Installation	4-1
4.1 Power and Grounding Cables	4-1
4.2 Installation Flowchart.....	4-1
4.3 Power and Grounding Cable Structure.....	4-2
4.4 Power and Grounding Cable Installation	4-4
4.4.1 19-Inch Power Distributor.....	4-4
4.4.2 -48 V Input Power Cable Installation	4-6
4.4.3 Internal Power Cable Installation	4-7
4.4.4 Grounding Cable Installation	4-8
Chapter 5 Cable Connection	5-1
5.1 Overall Cabling	5-1
5.2 Cable Layout	5-2
5.2.1 Cabling Principles	5-2
5.2.2 Cable Bundling.....	5-3
5.3 E1/T1 Cable.....	5-6
5.4 Monitoring Cables	5-10
5.4.1 Entrance Control Monitoring Cable	5-11
5.4.2 Smog Monitoring Cable	5-12
5.4.3 Flood Monitoring Cable.....	5-13
5.4.4 Temperature Monitoring Cable	5-14
5.4.5 Humidity Monitoring Cable	5-15
5.5 Serial Port Cable	5-16
5.6 Network Cable	5-18

5.7 Optical Fiber	5-19
Chapter 6 Installation Check	6-1
6.1 Overall Inspection	6-1
6.2 Power-ON/OFF Inspection.....	6-2
6.2.1 Inspection before Power-ON	6-2
6.2.2 Power-ON Process.....	6-3
6.2.3 Power-OFF Process	6-4
Appendix A Indoor Cabinet Grounding System Connection	A-1
A.1 Overview	A-1
A.2 Central Office Cabinet Grounding.....	A-1
A.2.1 Central Office Cabinet Grounding.....	A-1
A.2.2 Remote Office Cabinet Grounding.....	A-3
Appendix B Grounding Network Requirements	B-1
B.1 Grounding Cable Requirements	B-1
B.2 Cabinet and Central Office.....	B-2
B.3 Outgoing/Incoming Cables from/to Central Office	B-2
Appendix C Lightning Proof Network.....	C-1
C.1 Lightning Requirements	C-1
C.2 Lightning Design.....	C-1
C.3 Lightning Protection Grounding in Central Office	C-2
C.4 Lightning Protection Grounding in Remote Office	C-2
Figures.....	I
Tables	III
Glossary	V

About This Manual

Purpose

The ZXA10 C300 Optical Access Convergence Equipment (ZXA10 C300 for short) is a full-service optical access platform. It supports the video, data, voice, TDM, and CATV services. The ZXA10 C300 can be connected with MDUs/ONUs/ONTs through various networking technologies.

The ZXA10 C300 device consists of the switching and control cards, PON interface cards, uplink cards, Ethernet interface card, P2P interface cards, TDM interface cards, and common interface card. It provides large-capacity controllable multicast and high QoS control.

This manual provides procedures and guidelines that support the installation of the ZXA10 C300 Optical Access Convergence Equipment.

Intended Audience

This document is intended for the installation and debugging engineers.

Prerequisite Skill and Knowledge

To use this document effectively, users should have a general understanding of network technology. Familiarity with the following is helpful:

- The ZXA10 C300 system and its various components
- Maintenance procedures
- Local operating procedures

What Is in This Manual

This manual contains the following chapters:

Chapter	Summary
Chapter 1, Overview	Provides the device introduction, features, interfaces, configuration, components, standard compliance, technical specifications, and performance specifications.
Chapter 2, Pre-Installation Preparation	Describes the engineering documents preparation, tools and meters preparation, installation environment check, equipment room layout, and equipment unpacking.

Chapter	Summary
Chapter 3, Cabinet Installation	Describes the cabinet installation flow, cabinet installation, decorative panel installation, shelf installation, card installation, and fan tray installation.
Chapter 4, Power Supply Installation	Describes the power cable and grounding cable installation flow, cable details, and fan tray power supply system.
Chapter 5, Cable Connection	Describes the overall cabling, cabling principles, maintenance cables, and optical fiber layout.
Chapter 6, Installation Check	Describes the overall inspection, power supply inspection, and power-ON/OFF inspection.
Appendix A, Indoor Cabinet Grounding System Connection	Describes the grounding for rack system in central equipment room and remote equipment room.
Appendix B, Grounding Network Requirements	Describes the grounding requirements for the access network, grounding cable requirements, grounding connection requirements, grounding resistance requirements, ingress and egress communication office (station) cable requirements.
Appendix C, Lightning Proof Network	Describes the lightning protection requirements and lightning protection design.

Conventions

ZTE documents employ the following typographical conventions.

Typeface	Meaning
Italics	References to other Manuals and documents.
“Quotes”	Links on screens.
Bold	Menus, menu options, function names, input fields, radio button names, check boxes, drop-down lists, dialog box names, window names.
CAPS	Keys on the keyboard and buttons on screens and company name.
	Note: Provides additional information about a certain topic.
	Checkpoint: Indicates that a particular step needs to be checked before proceeding further.
	Tip: Indicates a suggestion or hint to make things easier or more productive for the reader.

Mouse operation conventions are listed as follows:

Typeface	Meaning
Click	Refers to clicking the primary mouse button (usually the left mouse button) once.
Double-click	Refers to quickly clicking the primary mouse button (usually the left mouse button) twice.
Right-click	Refers to clicking the secondary mouse button (usually the right mouse button) once.

Chapter 1

Overview

Table of Contents

Cabinet Introduction	1-1
Cabinet Features.....	1-2
Equipment Configuration	1-3
System Components	1-7
Technical Standards	1-8
Technical Specifications	1-10
Performance Specifications	1-11

1.1 Cabinet Introduction

According to different shelf sizes, ZXA10 C300 uses 19-inch indoor cabinets B6030-22C-IB and B6060-22F, and 21-inch indoor cabinet B6030-22C-EB. Each cabinet can hold two ZXA10 C300 shelves.

This manual introduces 19-inch cabinets.

- B6030-22C-IB cabinet
Dimensions: 2200 mm × 600 mm × 300 mm (Height × Width × Depth)
It is used to hold the 19-inch shelf (with the width of 19 inches, that is, 483 mm).
- B6060-22F cabinet
Dimensions: 2200 mm × 600 mm × 600 mm (Height × Width × Depth)
It is used to hold the 19-inch shelf.

Figure 1-1 shows the B6030-22C-IB cabinet appearance.

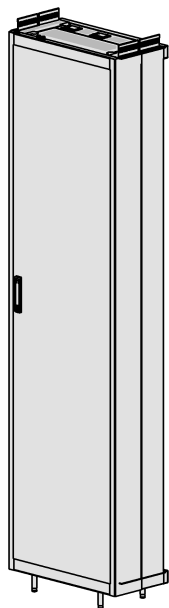
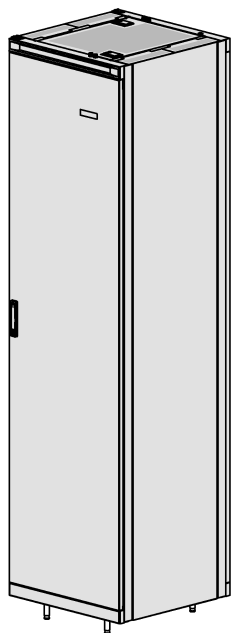
Figure 1-1 Cabinet Appearance

Figure 1-2 shows the B6060-22F cabinet appearance.

Figure 1-2 Cabinet Appearance

1.2 Cabinet Features

The B6030-22C-IB and B6060-22F cabinets are applicable to the areas where there are a large number of users and there are equipment rooms.

The cabinets have the following features:

- Large capacity

- Easy installation
- Easy cabling
- Complete monitoring functions
- High anti-interference capability
- Good ventilation
- Nice appearance

1.3 Equipment Configuration

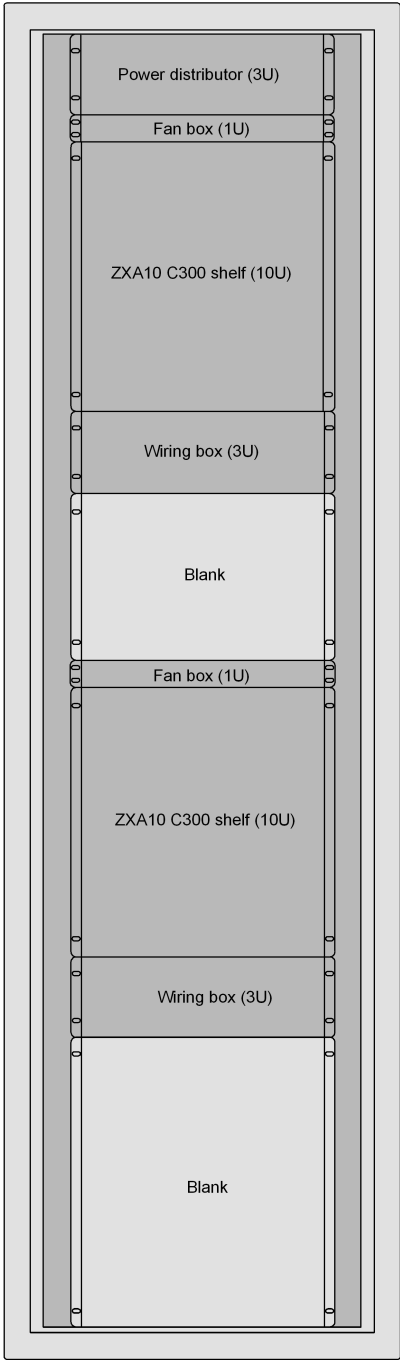
This section includes the following:

- Cabinet Configuration
- Shelf Configuration
- Cards

1.3.1 Cabinet Configuration

Figure 1-3 shows the cabinet configuration for the 19-inch shelves.

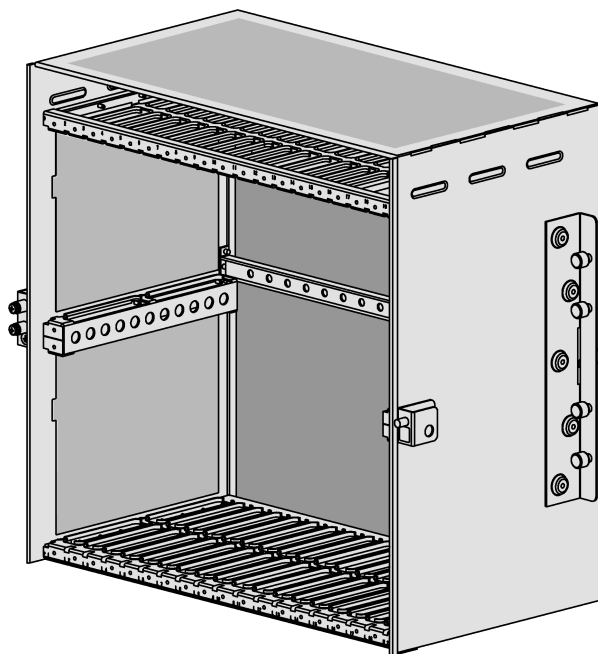
Figure 1-3 Cabinet Configuration for 19-Inch Shelves



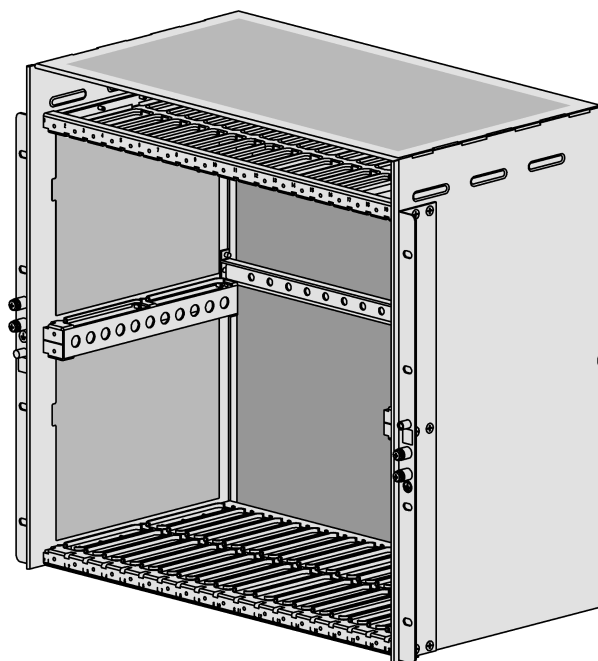
1.3.2 Shelf Configuration

Shelf Structure

When the 19-inch shelf is installed in a B6030-22C-IB cabinet, its structure is as shown in [Figure 1-4](#).

Figure 1-4 19-Inch Shelf Structure

When the 19-inch shelf is installed in a B6060-22F cabinet, its structure is as shown in Figure 1-5.

Figure 1-5 19-Inch Shelf Structure

Shelf Configuration

Figure 1-6 shows the 19-inch shelf configuration.

Figure 1-6 19-Inch Shelf Configuration

0	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
Power Card	Service Card	Service Card	Service Card	Service Card	Service Card	Service Card	Service Card	Service Card	Switching Control Card	Switching Control Card	Service Card	Service Card	Service Card	Service Card	Service Card	Common Interface Card	Uplink Card	Uplink Card
1																		
Power Card																		

Technical Specifications

Table 1-1 lists technical specifications of the 19-inch shelf.

Table 1-1 19-Inch Shelf Technical Specifications

Item	Specification
Dimensions	443.7 mm × 482.6 mm × 270 mm (Height × Width × Depth)
Weight	12.5 kg (Empty) 34.2 kg (Full configuration)

1.3.3 Cards

The ZXA10 C300 shelf consists of cards and a fan module, as listed in Table 1-2.

Table 1-2 Card List

Card Type	Name	Description
Switching and control card	SCXM	Type-M switching and control card
	SCXL	Type-L switching and control card
PON interface card	GTGO	8-port GPON interface card
	GTGQ	4-port GPON interface card
	ETGO	8-port EPON interface card
	ETXD	2-port 10G EPON asymmetric interface card

Card Type	Name	Description
Uplink card	XUTQ	4-port 10GE optical interface Ethernet uplink card
	GUFQ	4-port GE/FE optical interface Ethernet uplink card
	GUTQ	4-port GE electrical interface Ethernet uplink card
	GUSQ	2-port optical and 2-port electrical interface Ethernet uplink card
	HUTQ	2-port 10GE and 2-port GE optical interface Ethernet uplink card
	HUGQ	2-port GE and 2-port FE optical interface Ethernet uplink card
	HUTQ	2-port 10GE and 2-port GE optical interface Ethernet uplink card
	HUWQ	2-port 10GE and 2-port GE optical interface Ethernet uplink card
P2P interface card	FTGH	16-port P2P Ethernet interface card
	FTGK	48-port P2P Ethernet interface card
Ethernet interface card	GDFO	8-port GE optical Ethernet interface card
TDM interface card	CTLA	STM-N CES interface card
	CTBB	32-port E1 balanced CES interface card
	CTTB	32-port T1 CES interface card
	CTUB	32-port E1 unbalanced CES interface card
Common interface card	CICG	Common interface card
Backplane	MWIA	19-inch backplane
	MWIT	
	MWIR	
Power card	PRWG	4.5U power interface card
Fan module	FAN-C300	Fan module

1.4 System Components

The ZXA10 C300 cabinet consists of the following components:

Power Supply Module

The equipment room provides -48 V DC power for the cabinet.

The PRWG power card provides the secondary power to all the interface cards.

Fan Module

Fan module is a box, which is 1U in height and 19 inches in width.

Wiring Area

The wiring area holds an independent wiring plug-in box. It provides an area for wiring and a dust-proof net.

Monitoring Module

The monitoring module is the common interface card, which provides clock, management, monitoring, and N:1 protection interfaces.

Functional Card Area

The functional card area in the IEC 19-inch shelf has seventeen 9U slots and four 4.5U slots.

1.5 Technical Standards

Table 1-3 lists the technical standards that the B6030-22C-IB cabinet complies with.

Table 1-3 Technical Standards

Standard No.	Standard Name
YDN 065-1997	General Technical Specifications of Telephone Switching Equipment Issued by the Ministry of Posts and Telecommunications
YDN 099-1998	Optical Synchronous Transmission Network Technical System
YDN 108-1998	Technical Specifications of Consistency Test of V5.2 Interface
YDN 033-1997	Technical Requirements for Local Telecommunication Network Management
YDN 034.2-1997	ISDN Users - Network Interface Technical Specifications
YDN 086-1998	SDH Transmission Network Management Technology System
YDN 088-1998	Automatic Switching Telephone (Digital) Network Technology Mechanism
YD/T 950-1998	Technical Requirements and Test Methods for Telecommunication Switching Equipment Over-Voltage Over-Current Prevention

Standard No.	Standard Name
YD/T1007-1999	Distribution of Transmission Performance Indices in Access Network
YD/T1033-2000	Transmission Performance Index Series
YD/T1011-1999	Technical Requirements and Testing Methods for Digital Synchronous Network Node Slave Clock
YD/T1012-1999	Digital Synchronous Network Node Clock Series and Timing Characteristics
YD/T1061-2000	Technical Requirements for Transmitting IP LAPS on Synchronous Digital Hierarchy (SDH) (Similar to X.85)
YD/T1022-1999	Functions Requirements for Synchronous Digital Hierarchy (SDH) Equipment
YD/T1099-2001	Technical Specifications for Gigabit Ethernet Exchange
YD/T1141-2001	Test Methods for Gigabit Ethernet Exchange
YD/T1160-2001	Access Network Technical Requirement - Broadband Access Network Based on Ethernet Technology
YD/T1238-2002	Technical Requirements for Multi Services Transmission Based on
GB-T 6879-1995	Technologies and Test Methods for 2048 Kbps 30-channel PCM Multiplexing Equipment
GF002-9002	General Technical Specifications of Telephone Switching Equipment Issued by the Ministry of Posts and Telecommunications
GB4798.1-86	Electronic and Electric Products Application Environment Conditions, Storage
GB4798.2-84	Electronic and Electric Products Application Environment Conditions, Transportation
GIB-Z35-93	Component Derating Rules
GJB450-88	General Guideline for Equipment Research and Production Reliability
GJB-299A-91	Prediction Manual for Electric Equipment Reliability
ANSI/IEEE Std 802.1D	Media Access Control (MAC) Network Bridge (1998)

Standard No.	Standard Name
ITU-T G.703	Physical/Electrical Characteristics of Series Digital Interfaces
ITU-T G.704	Structure of All Synchronization Frames of Primary Group Rate and Secondary Group Rate Series
ITU-T G.960	Digital Segment for ISDN Basic Access
ITU-T G.961	Digital Transmission System Accessed to the Local Full Metal Line at ISDN Basic Rate
ITU-T G.962	Digital Segment Used for Accessing of ISDN 2048 kbps Primary Group Rate
ITU-T I.430	Specifications for Layer 1 of Network Interface - ISDN Basic Access Users
ITU-T I.431	Specifications for Layer 1 of ISDN Primary Group Rate Access Users - Network Interface

1.6 Technical Specifications

Dimensions and Weight

The cabinet dimensions and weight are as follows:

- B6030-22C-IB cabinet
Dimensions: 2200 mm × 600 mm × 300 mm (Height × Width × Depth)
Weight (empty): 61 kg
Weight (full configuration): 172 kg
- B6060-22F cabinet
Dimensions: 2200 mm × 600 mm × 600 mm (Height × Width × Depth)
Weight (empty): 110 kg
Weight (full configuration): 222 kg

Operation Environment

The equipment has the following operation environment requirements:

- Temperature: -5°C to 45°C
- Humidity: 5% - 95%
- Temperature change ratio: ≤ 3 °C/min

Power Supply

The equipment has the following power supply requirements:

- Rate voltage: -48 V DC
- Voltage range: -57 V DC to -41 V DC

Power Consumption

The equipment power consumption is as follows:

- Cabinet with full configuration: 2800 W
- A single shelf: 1400 W

1.7 Performance Specifications

Table 1-4 lists the ZXA10 C300 reliability and security specifications.

Table 1-4 Reliability and Security Specifications

Item	Specification
Static anti-interference	GB/T 17626-1998 (IEC 61000-4-2:1995) Grade 3 (contact 6 kV, air 8 kV)
Surge anti-interference	GB/T 17626.5-1998 (IEC 61000-4-5:1995)
Electric fast pulse group anti-interference	GB/T 17626.4-1998 (IEC 61000-4-4:1995)
Radio frequency electromagnetic field radiation anti-interference	GB/T 17626.3-1998 (IEC 61000-4-3:1995)
Radio frequency electromagnetic field conduction anti-interference	GB/T 17626.6-1998 (IEC 61000-4-6:1995)
Power supply down anti-interference	GB/T 17626.11-1998 (IEC 61000-4-11:1995)
Conduction transmission	GB 9254-1998 (Grade A ITE) (CISPR 22: 1997)
Radiation transmission	GB 9254-1998 (Grade A ITE) (CISPR 22: 1997)
Leak current to ground	≤ 3.5 mA
Anti-electricity strength	EN60950
Power cable bridging	ITU.T K 20
Power cable over-voltage	ITU.T K 20

This page intentionally left blank.

Chapter 2

Pre-Installation Preparation

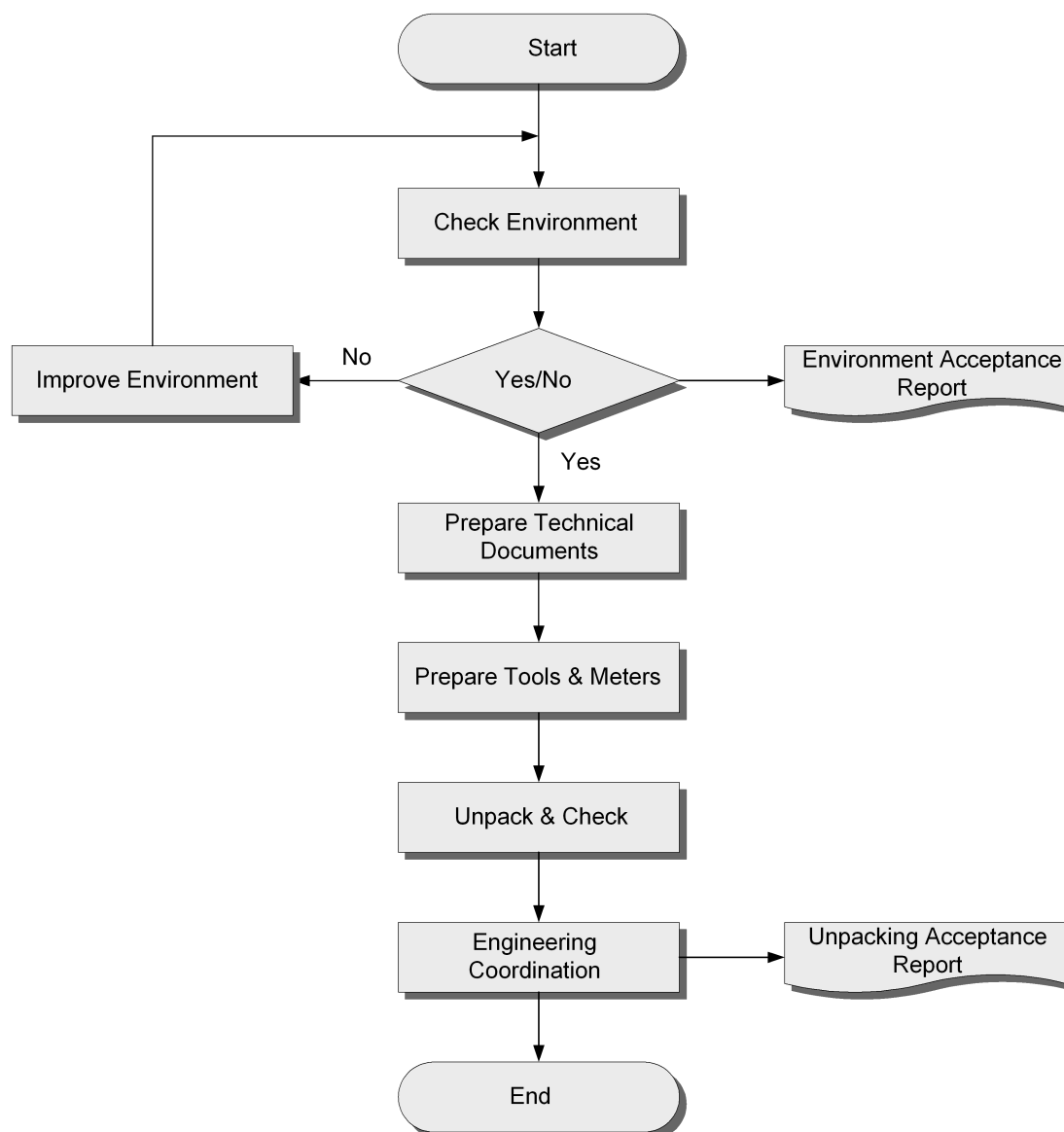
Table of Contents

Installation Preparation Flow	2-1
Engineering Documents Preparation	2-2
Tools and Meters	2-3
Installation Environment Check	2-4
Unpacking	2-7

2.1 Installation Preparation Flow

Figure 2-1 shows the installation preparation flow.

Figure 2-1 Installation Preparation Flow



2.2 Engineering Documents Preparation

The following documents are related to ZXA10 C300:

- Purchase contract (duplicate copy) and technical proposals.
- Engineering survey report, environment acceptance report, and installation acceptance report.
- ZXA10 C300 (V1.2) EPON/10G EPON Optical Access Convergence Equipment Hardware Installation Manual

2.3 Tools and Meters

General Tools and Meters

General tools and meters for installation are as follows:

- Tape
- Screw driver (straight, cross)
- Tweezers
- Spanner
- Diagonal pliers
- Cable peeler
- Sharp-nose pliers
- Cable clamp
- Unpacking pliers
- Electric iron
- Percussion drill
- Extraction tool
- Digital multimeter
- Thermometer
- Programming card
- Eraser
- Optical power meter
- Antistatic wrist strap
- Insulated tape
- Resistance tester
- Band
- Torque spanner

Special Tools and Meters

Special tools and meters for installation are as follows:

- Laptop
- Optical connector and pigtail fiber
- Optical attenuator



Tip:

All the tools and meters must be calibrated in advance. The general tools and meters are usually provided by the customer. ZTE installation engineers usually bring the special tools and meters with them.

2.4 Installation Environment Check

This topic includes the following:

- Construction Check
- Indoor Environment Check
- Power Supply Check
- Grounding Cable Check
- Security Check
- Other Supporting Facilities Check

2.4.1 Construction Check

Implement the following construction checks:

- Check civil work and make sure that indoor walls are dry.
- The equipment room must be large enough for holding all cabinets.
- The door must be greater than 3 meters in height and 1 meter in width.
- The load bearing capacity for the equipment floor must be more than 450 kg/m². The bearing capacity for the non-equipment floor must be at least 300 kg/m².
- Use air conditioners to keep the temperature and humidity within range.
- Take dust-proof measures to protect walls and roof.
- Lighting wires and power cables must be grounded and hidden.
- Make sure that no drain pipe passes through the equipment room.
- Label the location and measurement of the reserved hidden ducts. The hole size must be 400 mm × 400 mm. Ducts and troughs must be damp-proof.
- The anti-shock level must be up to seven.
- Mount lightning-arrester or rod to avoid lightning strike. Ground all metal pipes at the entrance of the equipment room.

2.4.2 Indoor Environment Check

Table 2-1 lists the operation environment requirements.

Table 2-1 Operation Environment Requirements

Parameter	Description
Ground resistance	< 1 ohm
Ambient temperature	- 5°C to 45°C
Ambient humidity	10% – 90%
Cleanness	Dust concentration with a diameter greater than 5 μm is equal to or less than 3 × 10 ⁴ grains/m ³ , and the dust is not electrically conductive, magnetically conductive, or erosive.

Parameter	Description
Illumination	- Do not expose the equipment room to direct sunshine, to prevent the circuit boards and other components from aging and deforming due to long-time exposure to the sunshine. - Average illumination is 150 lx - 200 lx. No glare. Normally, the fluorescence lamps embedded in the ceiling are used. - Use emergency lights in the equipment room.
Atmosphere pressure	70 kPa – 106 kPa
Air pollution	Do not expose the equipment to corrosive gases (such as H ₂ S, SO ₂ , oxides of SO ₂ and ammonia), smog or oil solvents. No smoking in the equipment room.

2.4.3 Power Supply Check

This topic describes the power supply requirements and power cable layout.

Power Supply Requirements

The B6030-22C-IB/B6030-22C-EB/B6060-22F cabinet uses –48 V **DC** power supply. The voltage is from -57 V DC to -41 V DC. The DC power supply must provide power-OFF protection measures and should be configured with storage batteries for backup.

Power Cable Layout

Power cable layout must meet the following requirements:

- The cabling routes, quantity, and position should be compliant with the general regulations of telecommunication projects. The specifications of the line conductor, insulation strength, and fuse capacity must meet the design requirements.
- The power cable must be one-piece cable without any joints.
- Positive and negative polarities of the DC power distribution box, busbar, and power cable must bear clear marks. The working ground wire end must be black. The negative polarity end must be blue. The protection earth end must be yellow and green.



Caution!

The power cables should be clearly separated from the signal lines and subscriber lines. In the process of laying out the power cables, make sure that the feed cables and busbar meet the medium-term or long-term expansion demands.

2.4.4 Grounding Cable Check

This topic describes the grounding requirements and factors affecting grounding.

Grounding Requirements

Grounding cables must meet the following requirements:

- Lay out the grounding cables in the equipment room in radial or plane mode. Use separate cables for the DC power distribution system protection grounding, working grounding, and lightning protection grounding.
- Do not form an electrical channel through reinforcing steel bars in buildings or through the cabinet.
- If the three grounding cables are unavailable separately at the installation site, combine the three grounding cables for grounding. The grounding resistance is less than 5 ohm.

Factors Affecting Grounding

The grounding resistance should be as little as possible. Factors that affect grounding resistance include resistance of grounding stake and connecting lead, contact resistance between grounding stake and soil, and soil type.

- The soil type has the greatest effect on grounding resistance. In an area with poor soil condition, it is recommended to add the resistance reducer (like acrylamide) around the grounding stub.
- Temperature also affects the grounding resistance. In northern China, it is recommended to embed stubs deep to reduce the temperature effect. The grounding stub must be made of galvanized material.
- Use electricity-conductive cables with cooper jacket to connect grounding stub and equipment. Their cross section area must be at least 50 mm² and the distance must be as short as possible. If necessary, give grounding connectors corrosive-proof to get a low resistance.

2.4.5 Security Check

Following are necessary security checks:

- Equip the room with fire extinguishers.
- Equip automatic fire control system for a large equipment room.
- Do not keep flammable or explosive materials in the equipment room.

2.4.6 Other Supporting Facilities Check

Purpose

Check whether other supporting facilities meet the equipment installation requirements.

Process

Based on the contract configuration requirements, check whether the uplink optical interface or electrical interface are available, whether the uplink devices are ready, whether the optical fibers are prepared, and whether the optical fiber connectors match. Conduct the following inspection based on the contract requirements:

- Check whether the power cables for external power supplies and cabinets are ready.
- Check whether the ODF, DDF, and MDF have been constructed.
- Check whether the optical fibers are prepared and whether the optical fiber connectors match.

2.5 Unpacking

This topic includes the following:

- Unpacking Cabinet
- Unpacking Shelf
- Unpacking Card



Note:

Only the system level packaging will be provided. The component or circuit pack packaging is unavailable.

2.5.1 Unpacking Cabinet

Perform this procedure to unpack and check cabinet.

Prerequisite

Before unpacking the cabinet, make sure that:

- The packing box is intact.
- The cabinet type is correct.

Context

To unpack and check the cabinet, perform the following steps:

Steps

1. Count the total number of the goods.

Before unpacking, it is necessary to count the number of the goods according to the packing list attached to the packing boxes. It is recommended to keep a record of goods.

2. Unpack the cabinet box.



Caution!

- Be cautious while unpacking. Protect the equipment coating and the circuit boards.
- When the equipment arrives at the destination with a considerably different temperature and humidity, do not open the package until it stays for some time in order to avoid damage to the equipment.
- Unpack carefully and check the items one by one. Make sure that the goods are consistent with the contract, shipping list, and packing list. Make sure that the component list and technical documents are in No.1 packing box.

- Unpack No.1 packing box and check whether the attached documents and unpacking inspection report are available. The installation staff must go through the technical documents and check the lists. Make detailed inspection and record if any damage of the interior packing is found.
- The cabinet is packed in a wooden box, as shown in [Figure 2-2](#). The cabinet is wrapped with aluminum foil and the edges are protected with cushions fixed with adhesive tape, as shown in [Figure 2-3](#).

Figure 2-2 Packing Box

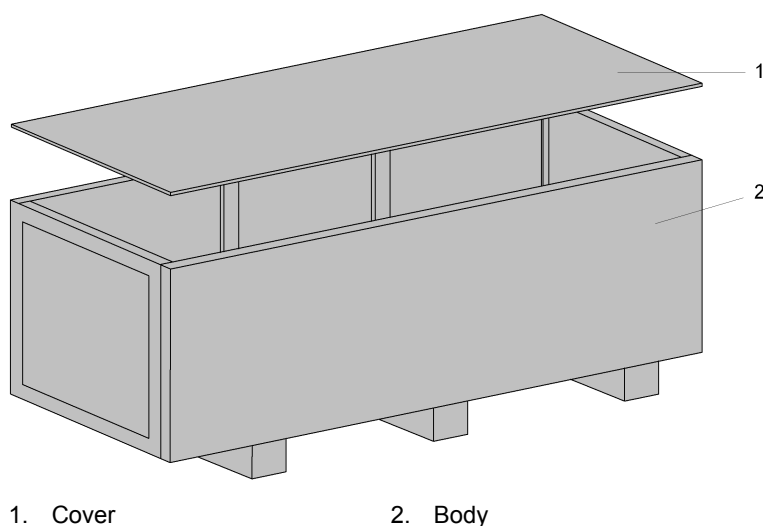
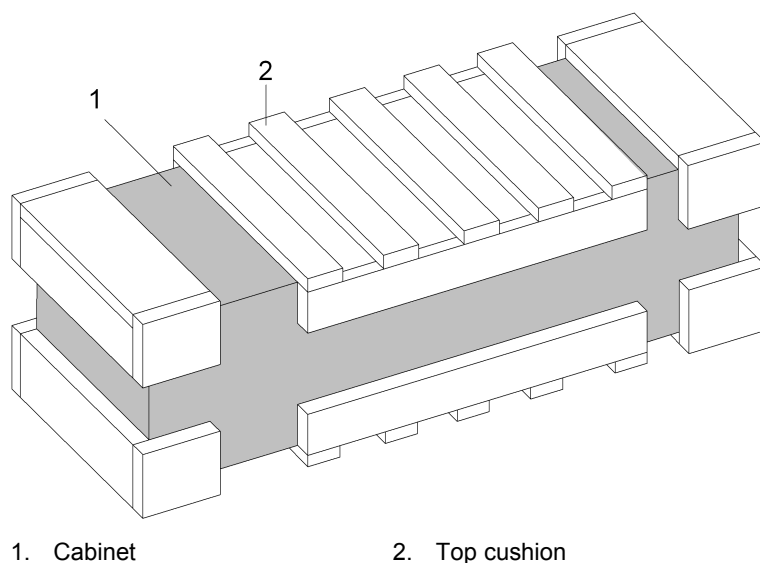


Figure 2-3 Cabinet Inner Package



Perform the following steps to open the box.

- a. Before unpacking, place the case horizontally upwards. Use unpacking pliers or nail claw to remove iron corner-wrappers of the box and the nails in the cover. The unpacking sequence is: Detach the top cover > Remove the front side plate > Remove the side plates (two sides) > Remove the back side plate > Remove the foam plates > Open the moisture-proof plastic film.

**Caution!**

Open the box carefully to avoid any damage to the cabinet.

- b. Remove the packing material such as foam cushion and aluminum foil bag on cabinet. Hold the lower part of the wooden box and pull the cabinet slowly till it is upright. Then, drag the box out slowly until it stands steady.

**Note:**

Make sure that at least five persons are present to handle this procedure.

- c. Keys required for cabinet are packed in a plastic bag and hung on the handle of the cabinet front door. Use the keys to open the front, rear, and side doors. Check whether there are complete collection of plug-in units in the cabinet. Check for cabinet cleanness, scratches, or loosened parts on the surface. Make sure that cabinet interior is clean with reliable socket connection and clear markings. Because the plug-in units are transported with the cabinet, their reinforcing mode,

installation accessories, and reliability should be checked carefully. Close the cabinet door when completing the cabinet inspection.

– End of Steps –

Result

The cabinet is successfully unpacked.

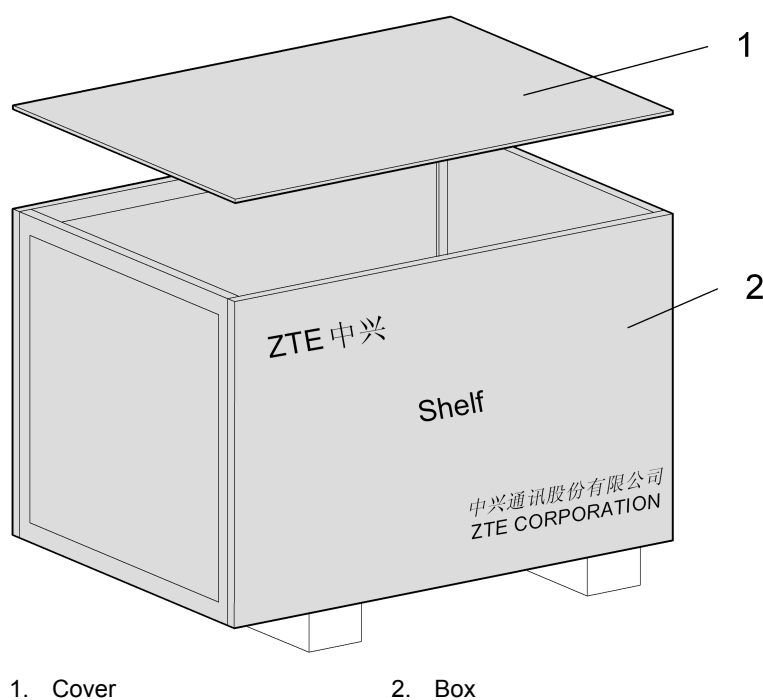
2.5.2 Unpacking Shelf

Perform this procedure to unpack the shelf.

Context

The shelf for expansion is packed separately in a box. [Figure 2-4](#) shows the shelf packing box.

Figure 2-4 Shelf Packing Box



To unpack the shelf, perform the following steps:

Steps

1. Place the packed unit on ground and keep its face upwards.
2. Use a nail hammer to remove nails and corner wrap sheets.
3. Remove the top cover of the packing box.

– End of Steps –

Result

The shelf is unpacked successfully.

Follow-Up Action

Fill in the acceptance list and the acceptance report. Both parties should sign for confirmation and each party should keep a copy.

Contact the local ZTE office, if anything is found damaged, wrong, or missing during the unpacking inspection.

2.5.3 Unpacking Cards

Perform this procedure to unpack cards.

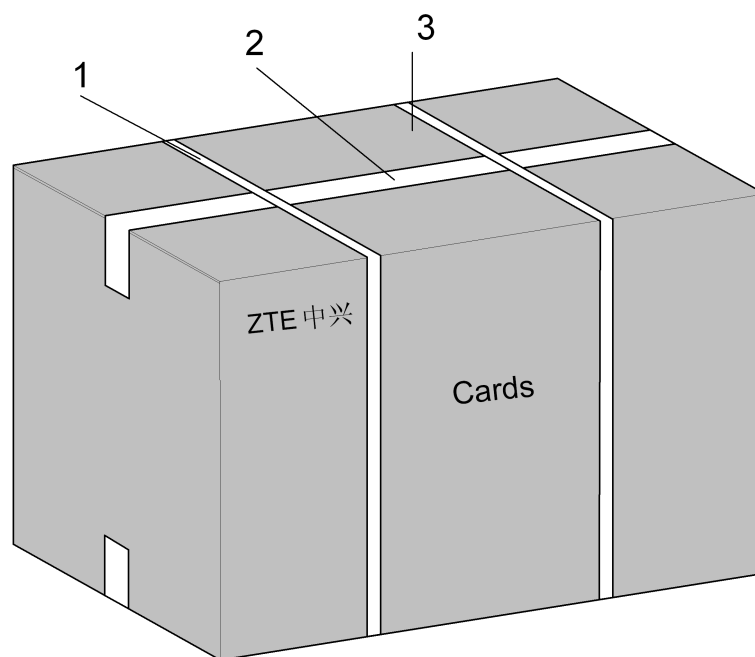
Context

To unpack a card, perform the following steps:

Steps

1. Unpack the card packing box, which is as shown in [Figure 2-5](#).

Figure 2-5 Card Packing Box

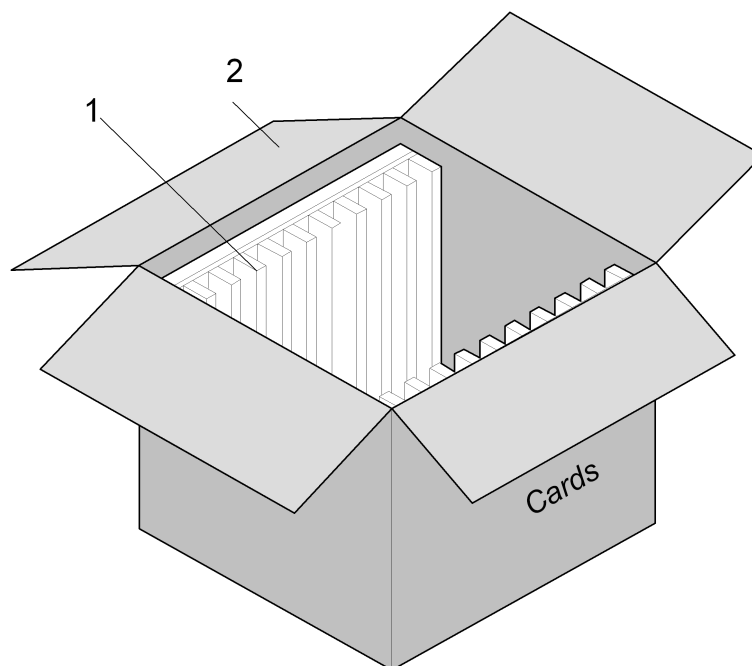


1. Strap

2. Seal tape

3. Carton

2. Open the card-packing unit slowly. [Figure 2-6](#) shows an opened card packing box.

Figure 2-6 Opened Card Packing Box

1. Card pad

2. Carton

3. Count the cards.
4. Wear the antistatic wrist straps and take the card from the antistatic bag. Conform the card type.
5. Check connectors, fiber tails, and accessories according to specifications and requirements.
6. Check the cable type, length, and cable connector.

– End of Steps –

Result

The cards are unpacked successfully.

Follow-Up Action

Fill in the acceptance list and the acceptance report. Both parties should sign it for confirmation and each party should keep a copy.

Contact the local ZTE office, if anything is found damaged, wrong, or missing during the unpacking inspection.

Chapter 3

Cabinet Installation

Table of Contents

Cabinet Installation Flowchart.....	3-1
Base Installation Mode	3-2
Foot Installation Mode	3-9
Cabinets Connection	3-14
Shelf Installation	3-19
Plug-in Box Installation	3-20
Card Installation	3-21

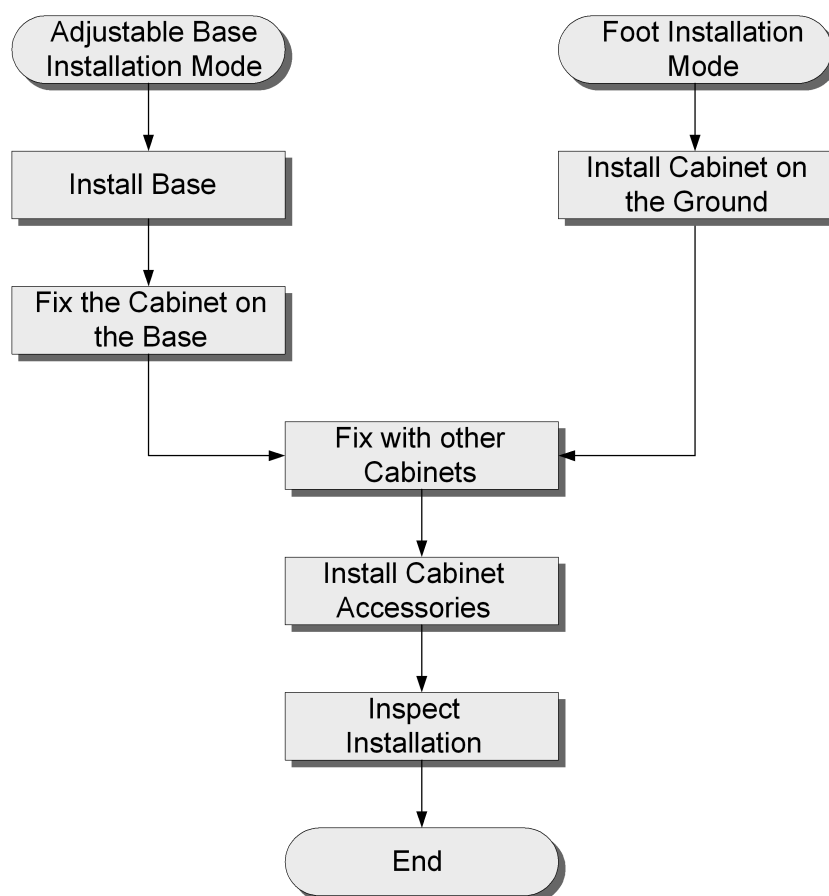
3.1 Cabinet Installation Flowchart

Confirm the location of the cabinet according to the *ZXA10 C300 (V1.2) Optical Access Convergence Equipment Project Survey Report* before installing the cabinet.

The cabinet can be installed in the following modes:

- Base installation mode
- Foot installation mode

Figure 3-1 shows the cabinet installation flowchart.

Figure 3-1 Cabinet Installation Flowchart

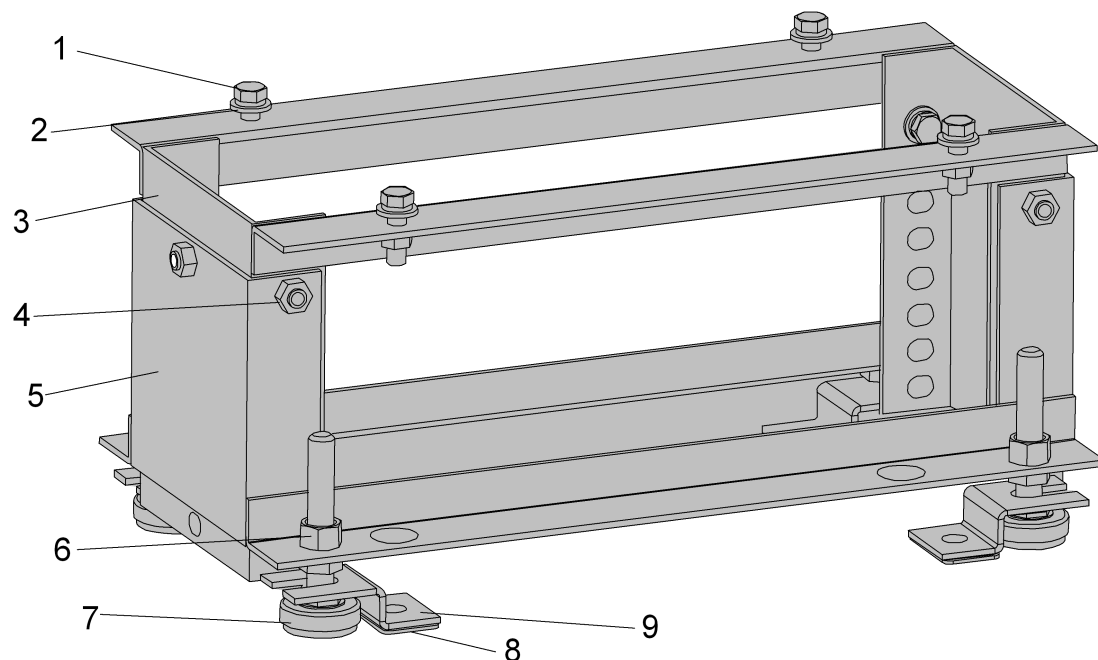
3.2 Base Installation Mode

Base installation mode is to install the cabinet with a base. This topic includes the following:

- Base Structure
- Base Installation Flowchart
- Positioning Base
- Installing Base
- Fixing Cabinet

3.2.1 Base Structure

Figure 3-2 shows the base structure of the B6030-22C-IB/B6030-22C-EB/B6060-22F cabinet.

Figure 3-2 Cabinet Base Structure

- | | | |
|---|----------------------------|-------------------|
| 1. Hex head screw M12 × 40
(with spring pad) | 4. Hex head screw M12 × 25 | 8. Insulation pad |
| 2. Washer | 5. Lower shelf | 9. Pressing plate |
| 3. Upper shelf | 6. Adjustable nut M16 | |
| | 7. Foot M16 × 90 | |

The base height is adjustable. There are three types of base according to the base height:

- Adjustable height from 160 mm to 260 mm
- Adjustable height from 260 mm to 460 mm
- Adjustable height from 460 mm to 660 mm

Figure 3-3 shows the base whose height is adjustable from 160 mm to 260 mm.

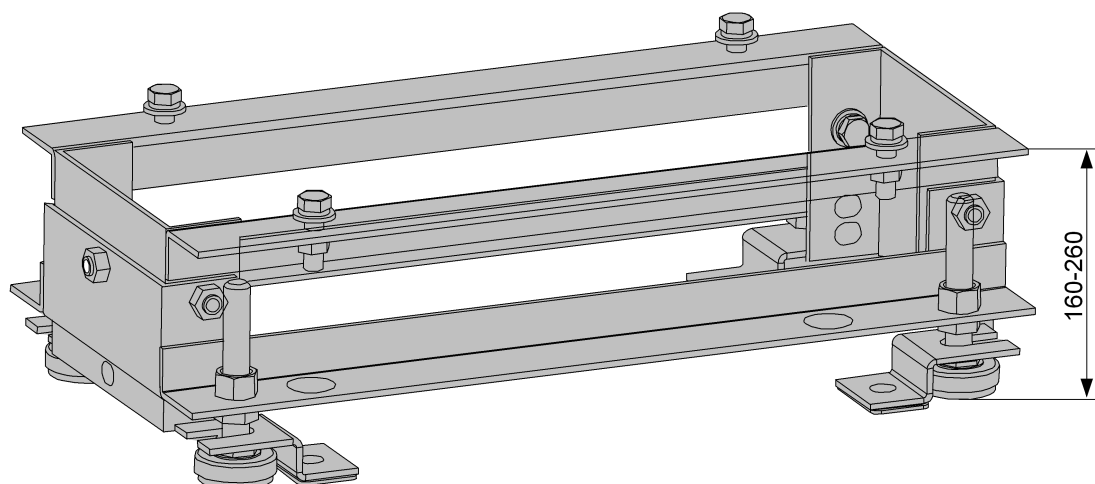
Figure 3-3 Base With Adjustable Height From 160 mm to 260 mm

Figure 3-4 shows the base whose height is adjustable from 260 mm to 460 mm.

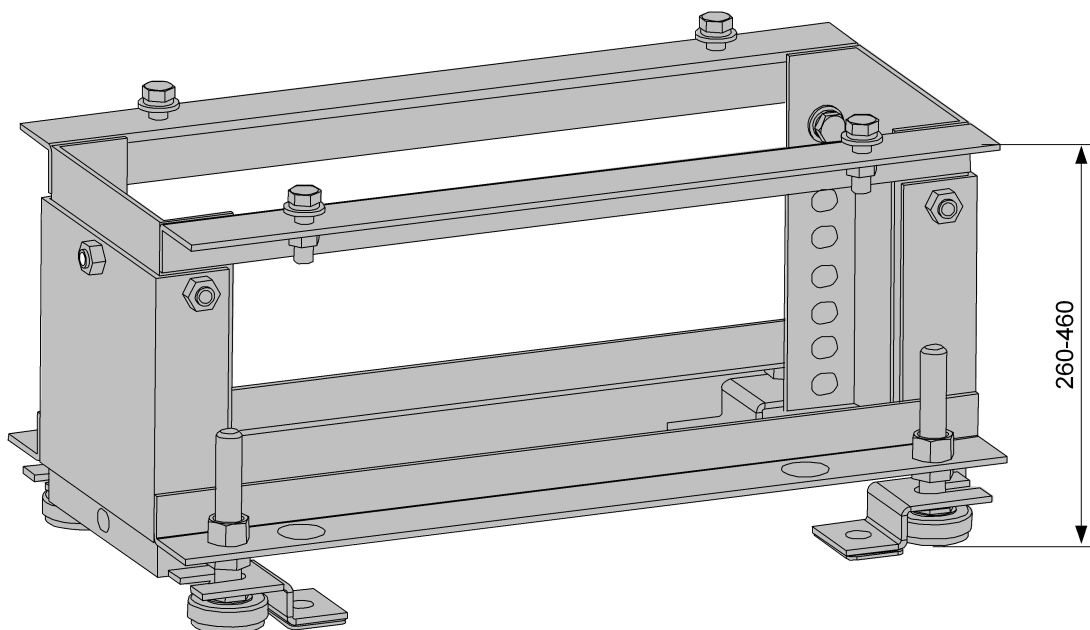
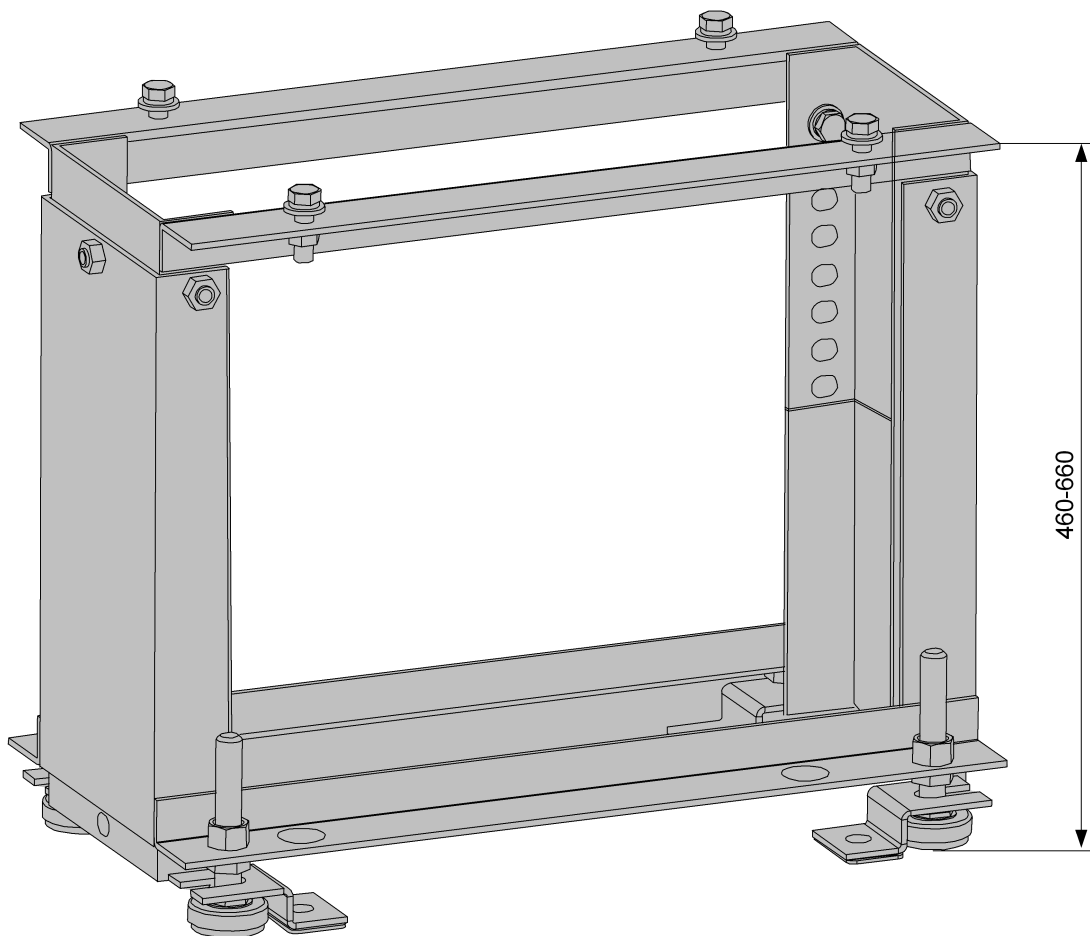
Figure 3-4 Base With Adjustable Height From 260 mm to 460 mm

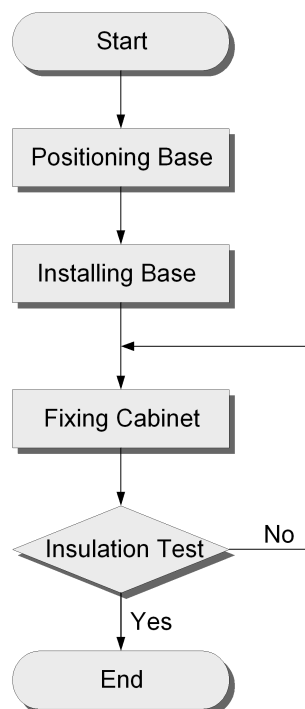
Figure 3-5 shows the base whose height is adjustable from 460 mm to 660 mm.

Figure 3-5 Base With Adjustable Height From 460 mm to 660 mm

3.2.2 Base Installation Flowchart

Figure 3-6 shows the base installation flowchart.

Figure 3-6 Base Installation Flowchart



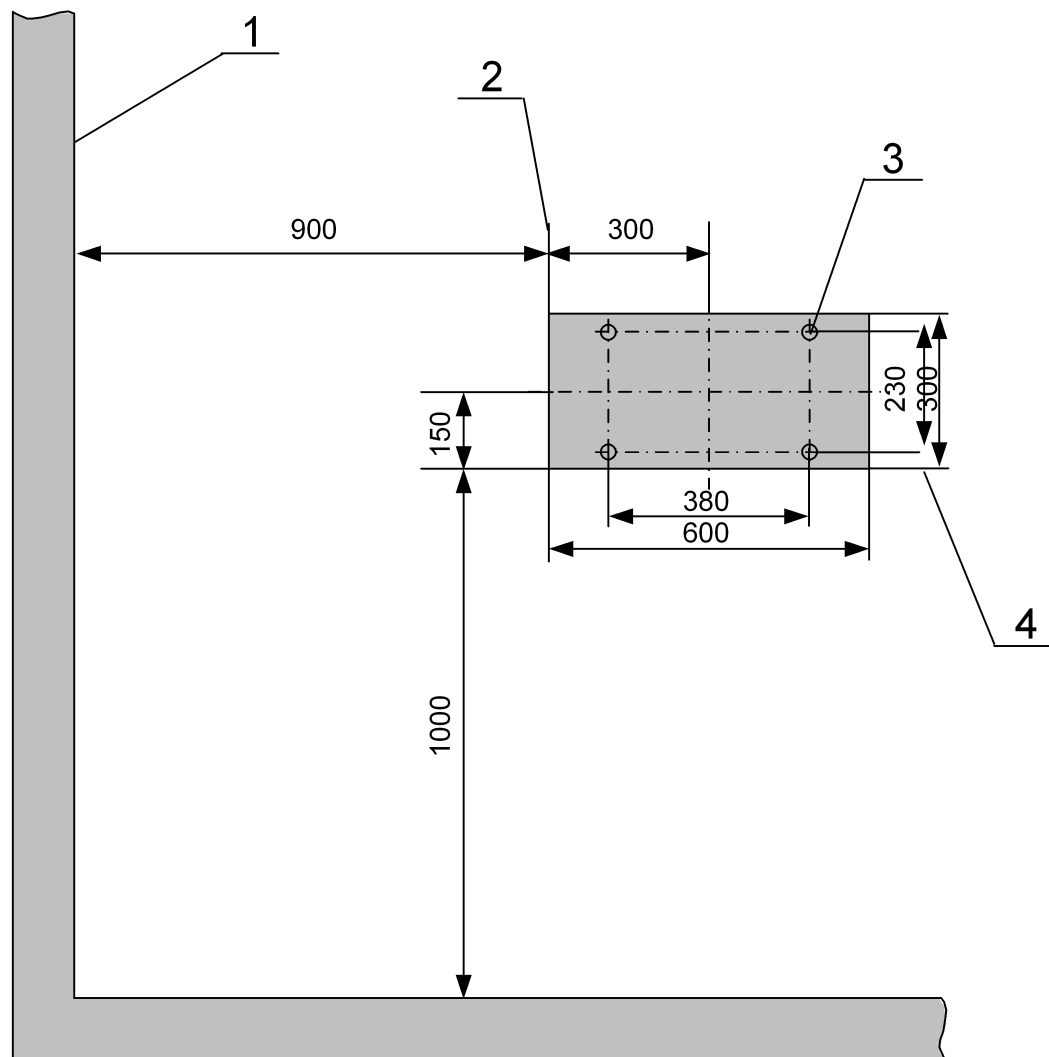
3.2.3 Positioning Base

Two steps are required for positioning the base.

1. Marking
 - a. Confirm the position for installing the cabinet according to the reference dimensions and base dimensions given in the engineering design diagram.
 - b. Use a measuring tape to measure the position.
 - c. Use an ink marker to mark the position.

Figure 3-7 shows the drilling diagram.

Figure 3-7 Drilling Diagram



2. Drilling

- Drill holes at the positions where the expansion bolt holes are marked.
- Keep the drill bit vertical to the ground and tightly hold the handle.
- Keep the depth of holes equal to the length of expansion bush plus cone.
- Keep the depths of all holes equal.
- While drilling, use a cleaner to remove dust.



Caution!

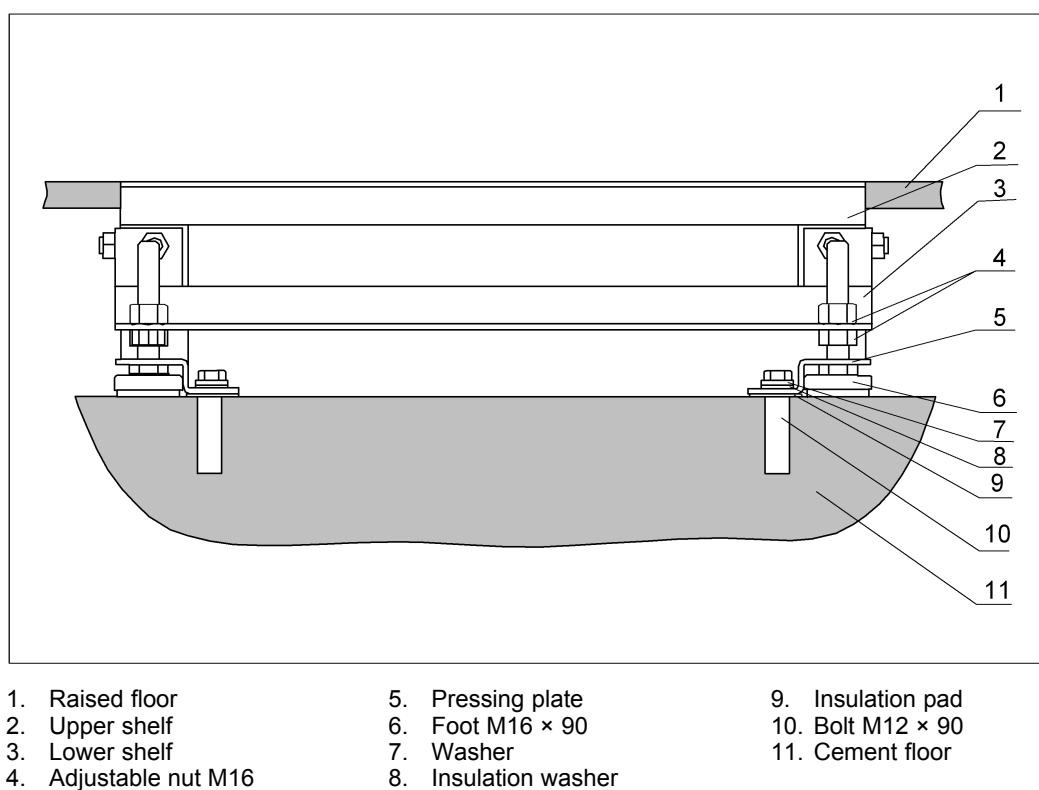
Marking and drilling are important for hardware installation. Improper marking and drilling may cause much inconvenience in the future work. The quality of marking and drilling is the prerequisite to the whole project.

3.2.4 Installing Base

The following steps are required for installing the base:

1. Clean holes with a cleaner.
2. Measure the space between holes. Reposition and re-drill the holes, which are largely deviated from the requirements.
3. Put the expansion bolt and expansion tube into the hole vertically.
4. Strike the expansion bolt with a rubber hammer until the expansion tube enters the ground completely. (The expansion bolt is M12 × 100, attached with the base.)
5. Install the base on the floor, as shown in [Figure 3-8](#).

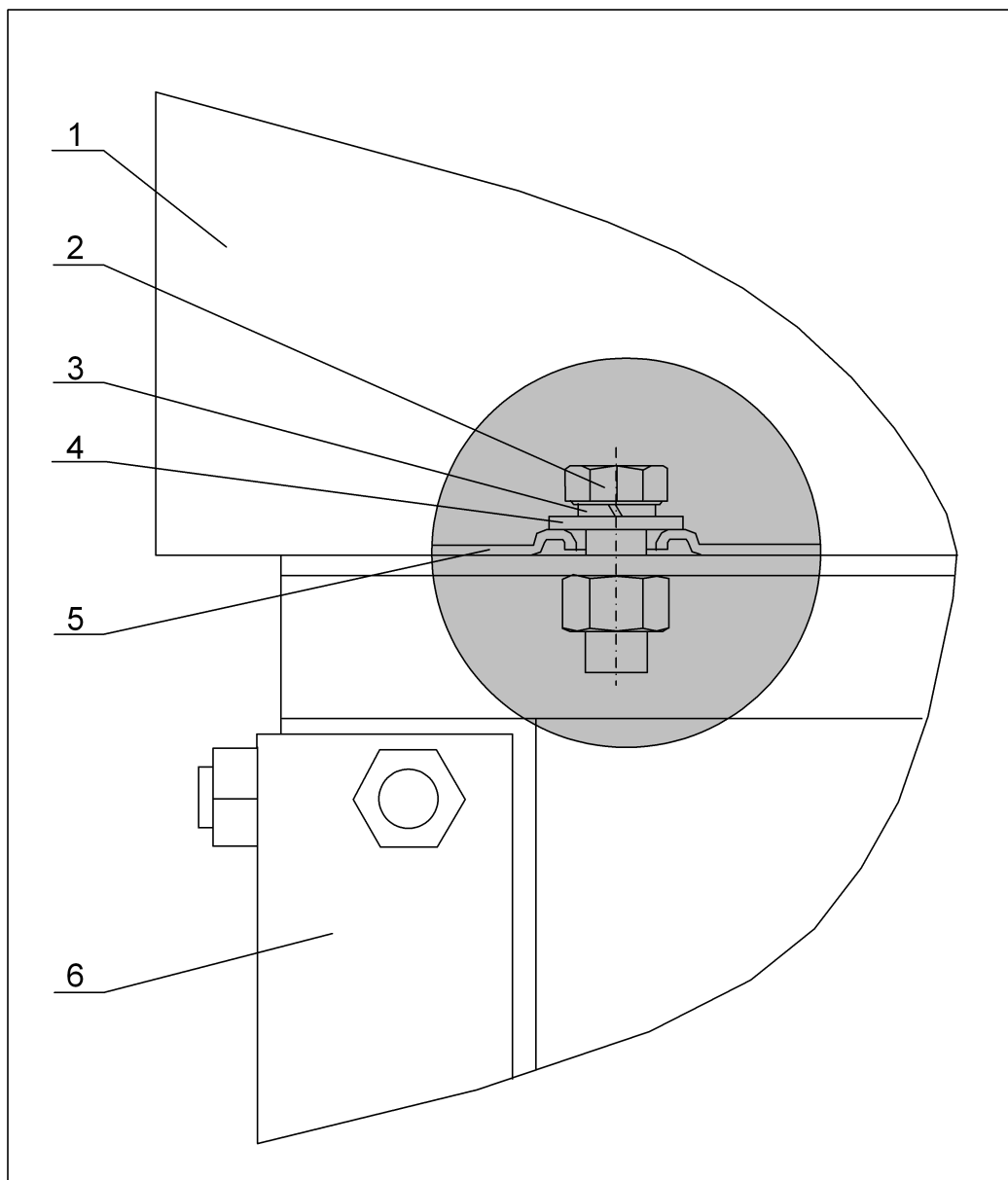
Figure 3-8 Installing Base



3.2.5 Fixing Cabinet

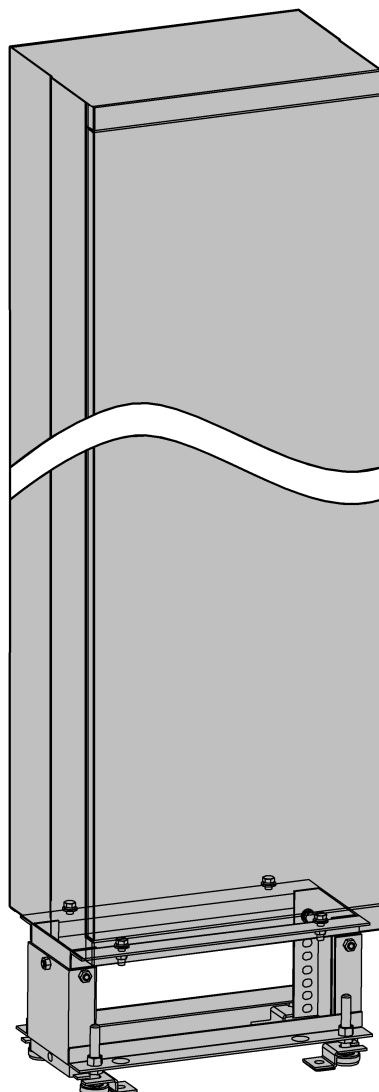
The following steps are required for fixing a cabinet:

1. Move the cabinet on to the base.
2. Use the foot under the base to adjust the level.
3. Fix the cabinet on the base, as shown in [Figure 3-9](#).

Figure 3-9 Fixing Cabinet

- | | | |
|-------------------------------|---------------------|--------------------|
| 1. Cabinet | 3. Spring washer 12 | 6. Adjustable base |
| 2. Hex head screw M12
× 40 | 4. Washer | |
| | 5. Cabinet bottom | |

4. [Figure 3-10](#) shows the cabinet that is successfully installed on the base.

Figure 3-10 Cabinet Installed on the Base

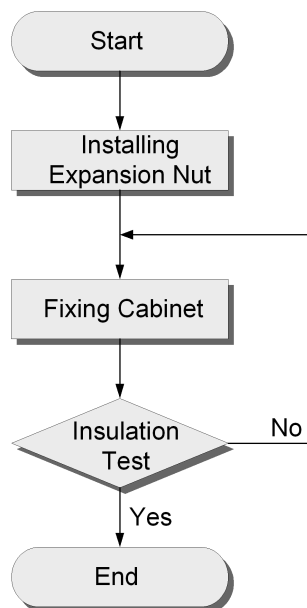
3.3 Foot Installation Mode

Foot installation mode is to install the cabinet on the ground directly. This topic includes the following:

- Foot Installation Flowchart
- Positioning Cabinet
- Installing Expansion Bolt
- Fixing Cabinet

3.3.1 Foot Installation Flowchart

Figure 3-11 shows the foot installation flowchart.

Figure 3-11 Foot Installation Flowchart

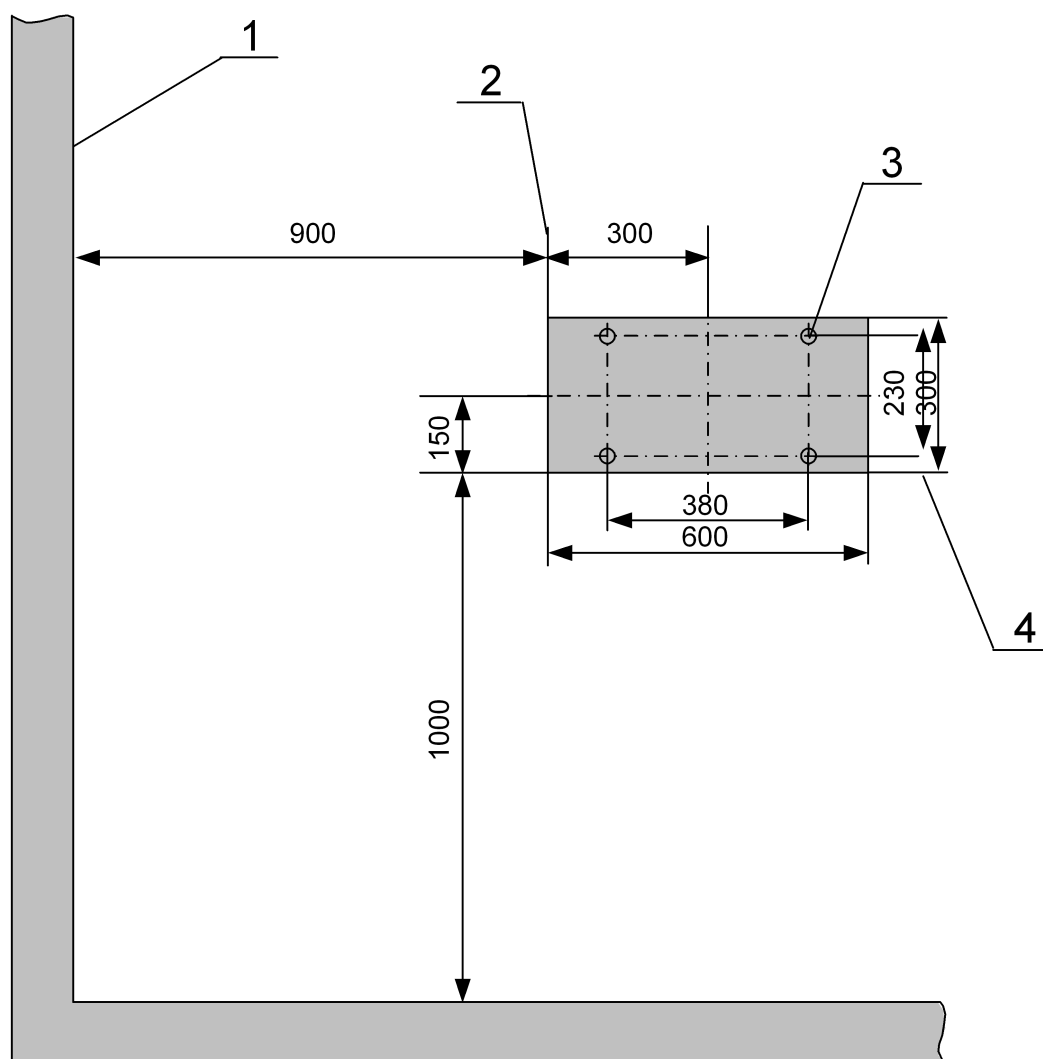
3.3.2 Positioning Cabinet

Two steps are required for positioning the cabinet.

1. Marking
 - a. Confirm the position for installing the cabinet according to the reference dimensions and base dimensions given in the engineering design diagram.
 - b. Use a measuring tape to measure the position.
 - c. Use an ink marker to mark the position.

Figure 3-12 shows the drilling diagram.

Figure 3-12 Drilling Diagram



2. Drilling

- Drill holes at the positions where the expansion bolt holes are marked.
- Keep the drill bit vertical to the ground and tightly hold the handle.
- Keep the depth of holes equal to the length of expansion bush plus cone.
- Keep the depths of all holes equal.
- While drilling, use a cleaner to remove dust.
- If the surface is too smooth and solid to locate the drill bit, cut a concave with a punching bit to help position the drill bit.

**Caution!**

Marking and drilling are important for hardware installation. Improper marking and drilling may cause much inconvenience in the future work. The quality of marking and drilling is the prerequisite to the whole project.

3.3.3 Installing Expansion Bolt

The following steps are required for installing an expansion bolt:

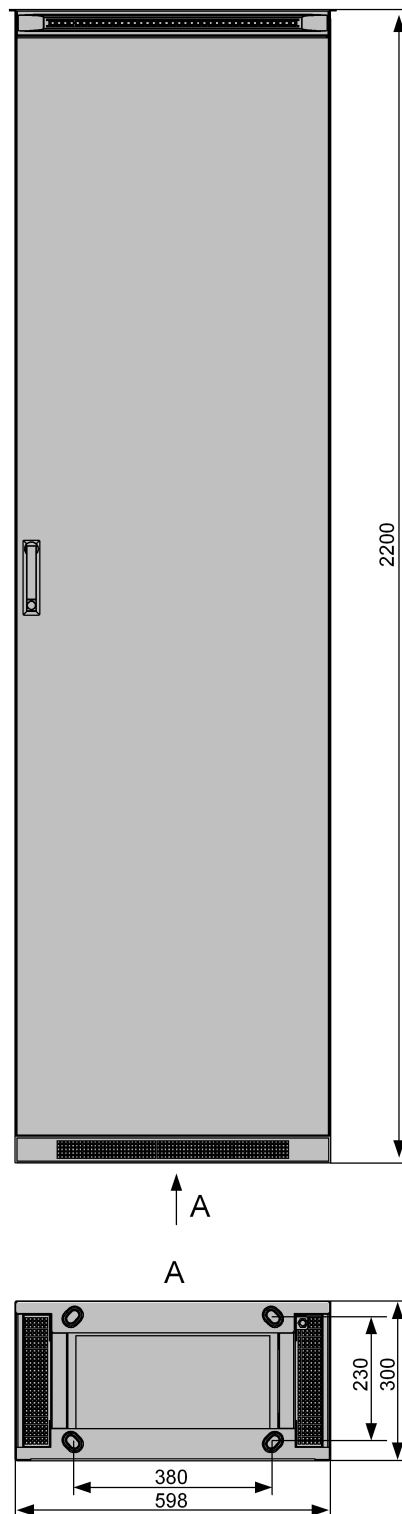
1. Clean the holes with a cleaner.
2. Measure the space between holes. Reposition and re-drill the holes, which are largely deviated from the requirements.
3. Put the expansion bolt and expansion tube into the hole vertically.
4. Strike the expansion bolt with a rubber hammer until the expansion tube enters the ground completely.

3.3.4 Fixing Cabinet

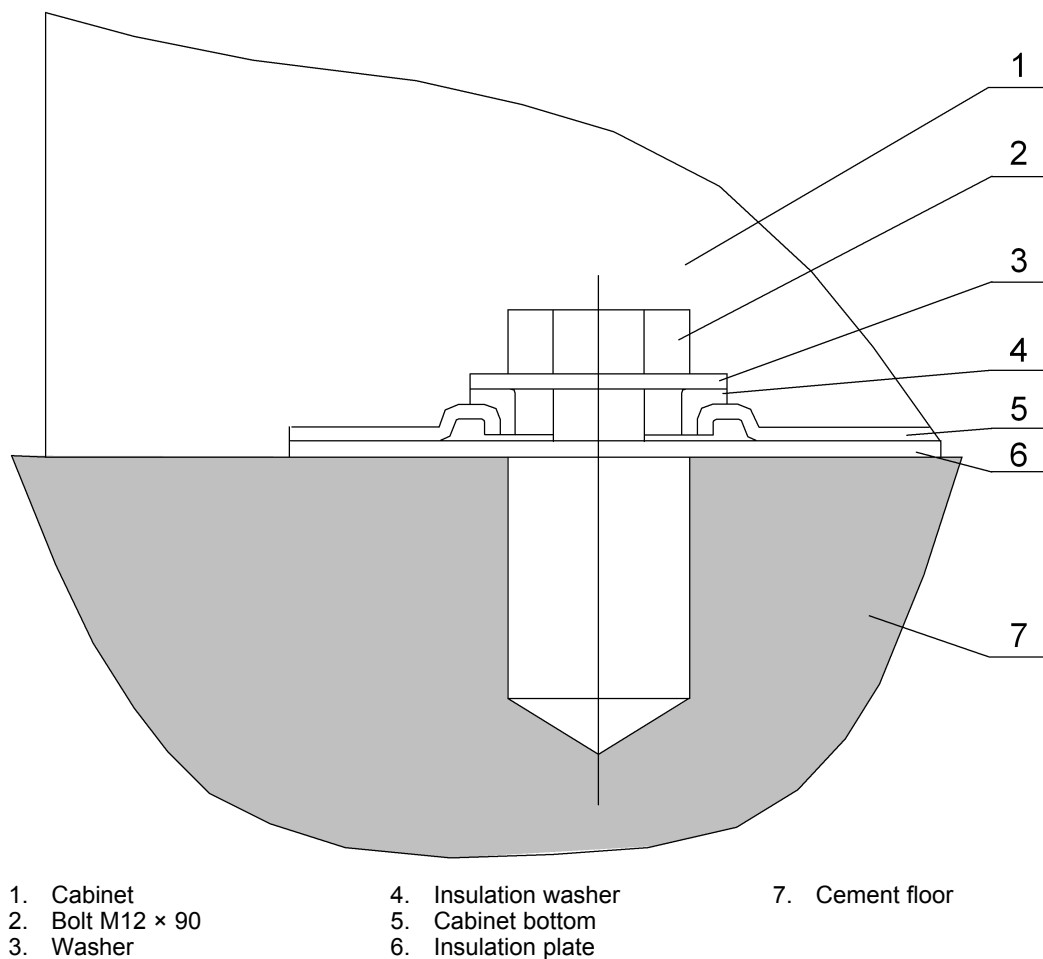
The following steps are required for fixing a cabinet on the cement floor:

1. Move the cabinet and put the four installation feet on the bottom of the cabinet into the holes drilled on the ground, as shown in [Figure 3-13](#).

Figure 3-13 Cabinet Installation Holes



2. Fix the cabinet on the cement floor, as shown in [Figure 3-14](#).

Figure 3-14 Fixing Cabinet

3.4 Cabinets Connection

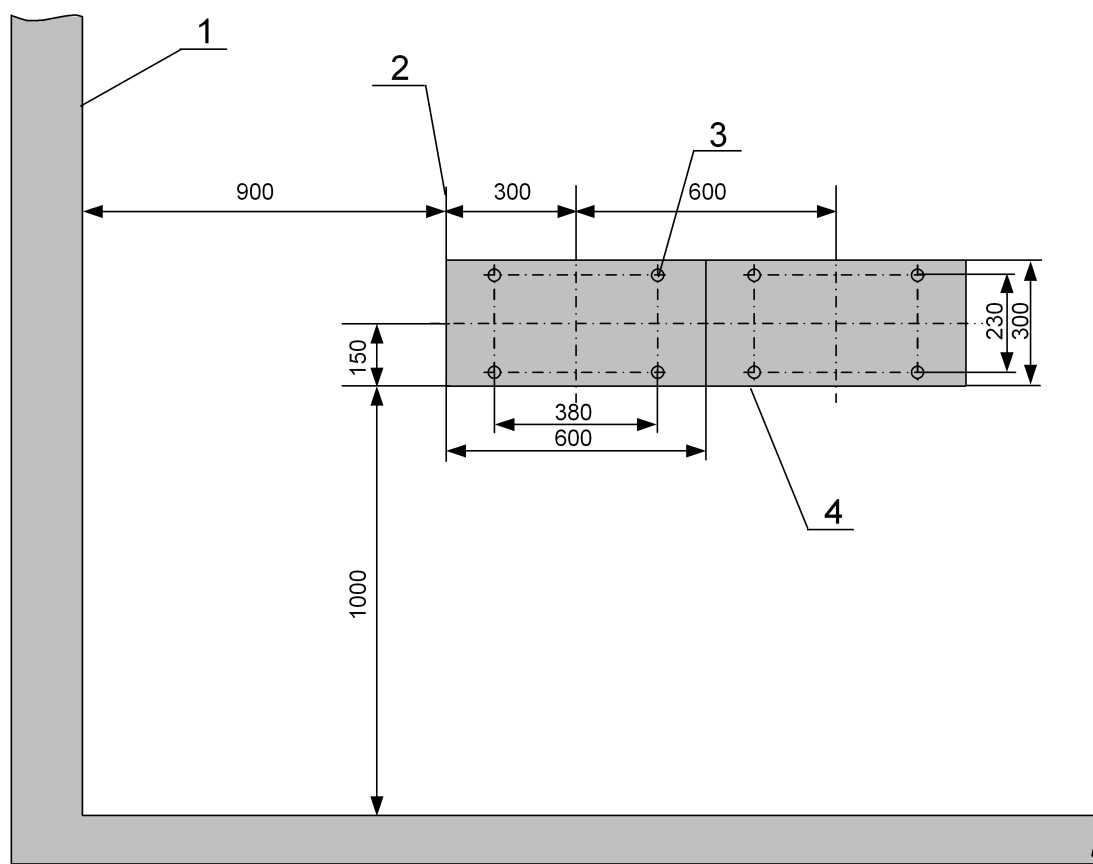
This topic includes the following:

- Cabinets Connection in Base Installation Mode
- Cabinets Connection in Foot Installation Mode
- Adjacent Cabinets Connection

3.4.1 Cabinets Connection in Base Installation Mode

Fixing Two Cabinets

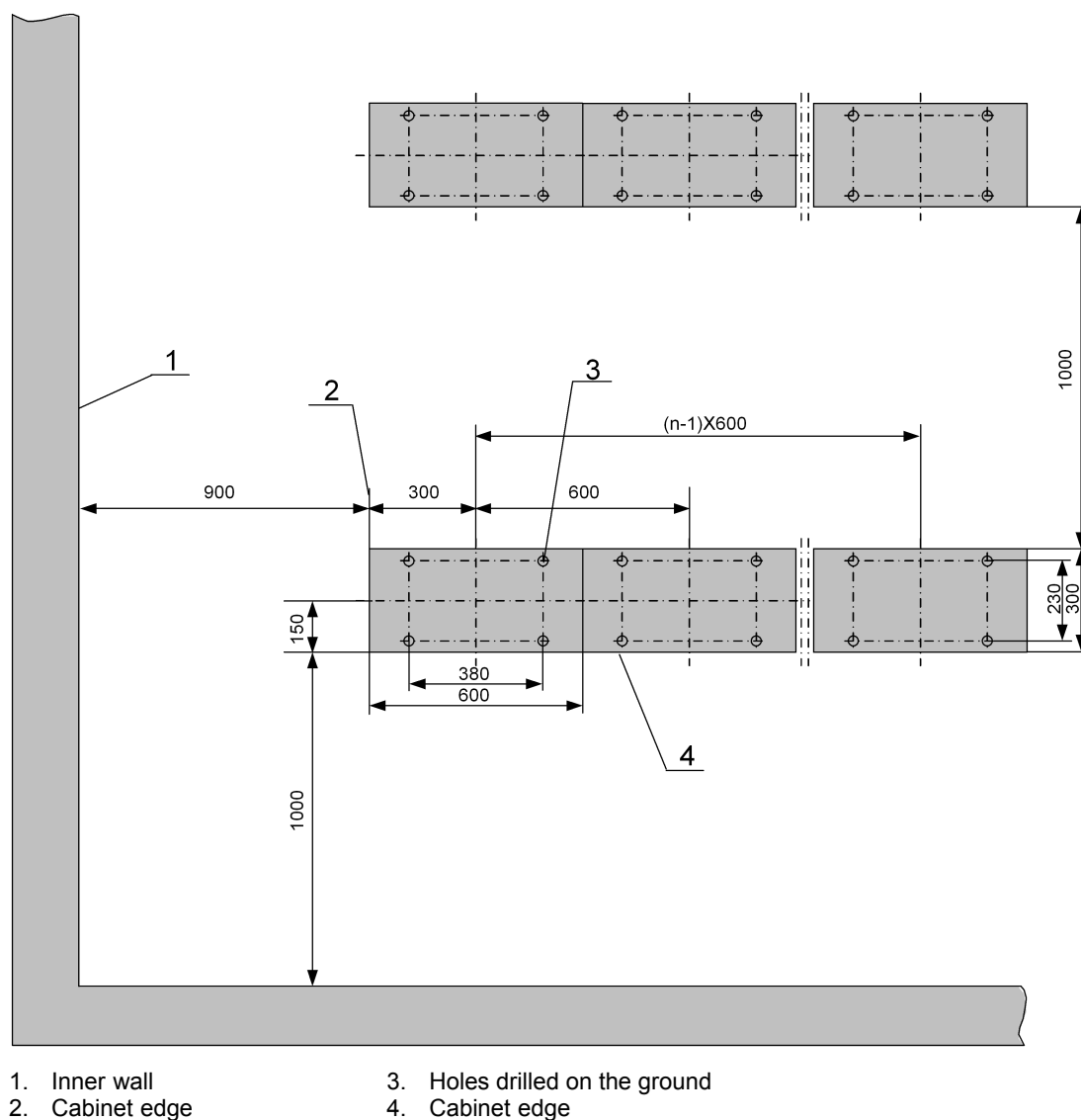
Fix two cabinets in parallel, as shown in [Figure 3-15](#).

Figure 3-15 Fixing Two Cabinets In Parallel

- | | |
|-----------------|--------------------------------|
| 1. Inner wall | 3. Holes drilled on the ground |
| 2. Cabinet edge | 4. Cabinet edge |

Fixing Multiple Cabinets

Fix multiple cabinets in parallel, as shown in [Figure 3-16](#).

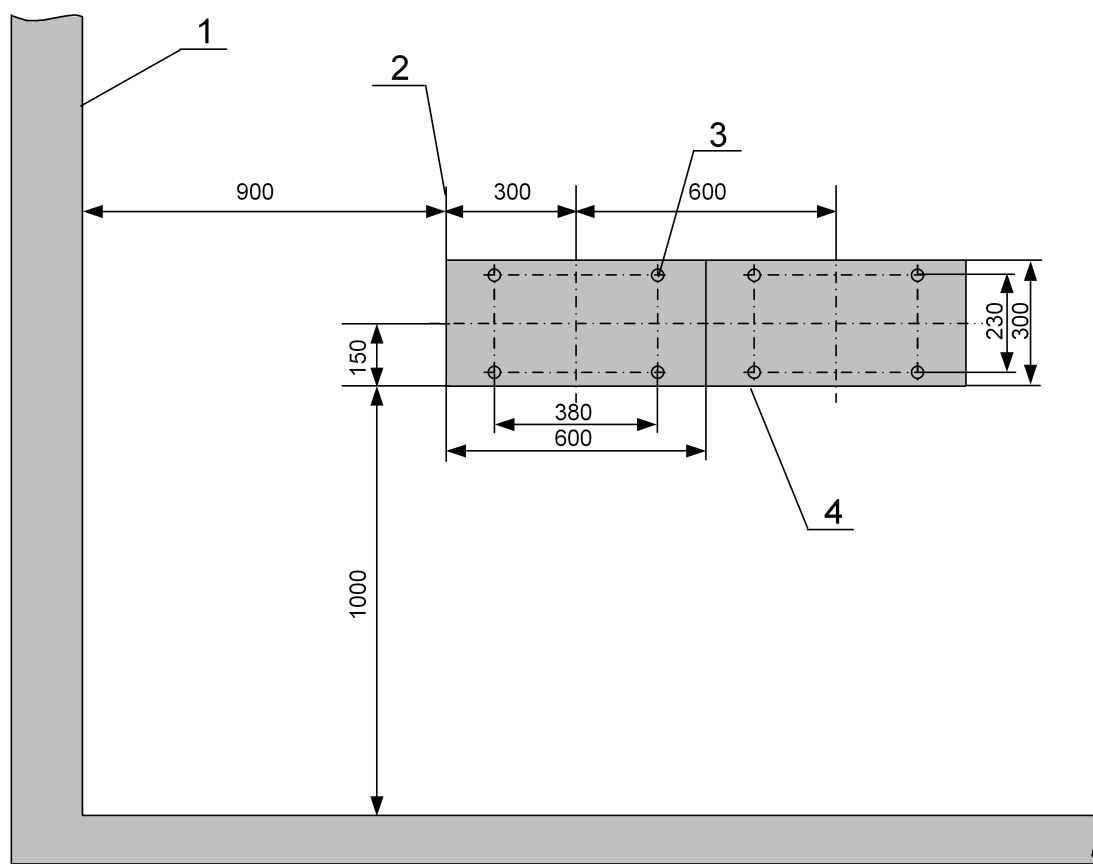
Figure 3-16 Fixing Multiple Cabinets In Parallel

3.4.2 Cabinets Connection in Foot Installation Mode

Fixing Two Cabinets

Fix two cabinets in parallel, as shown in [Figure 3-17](#).

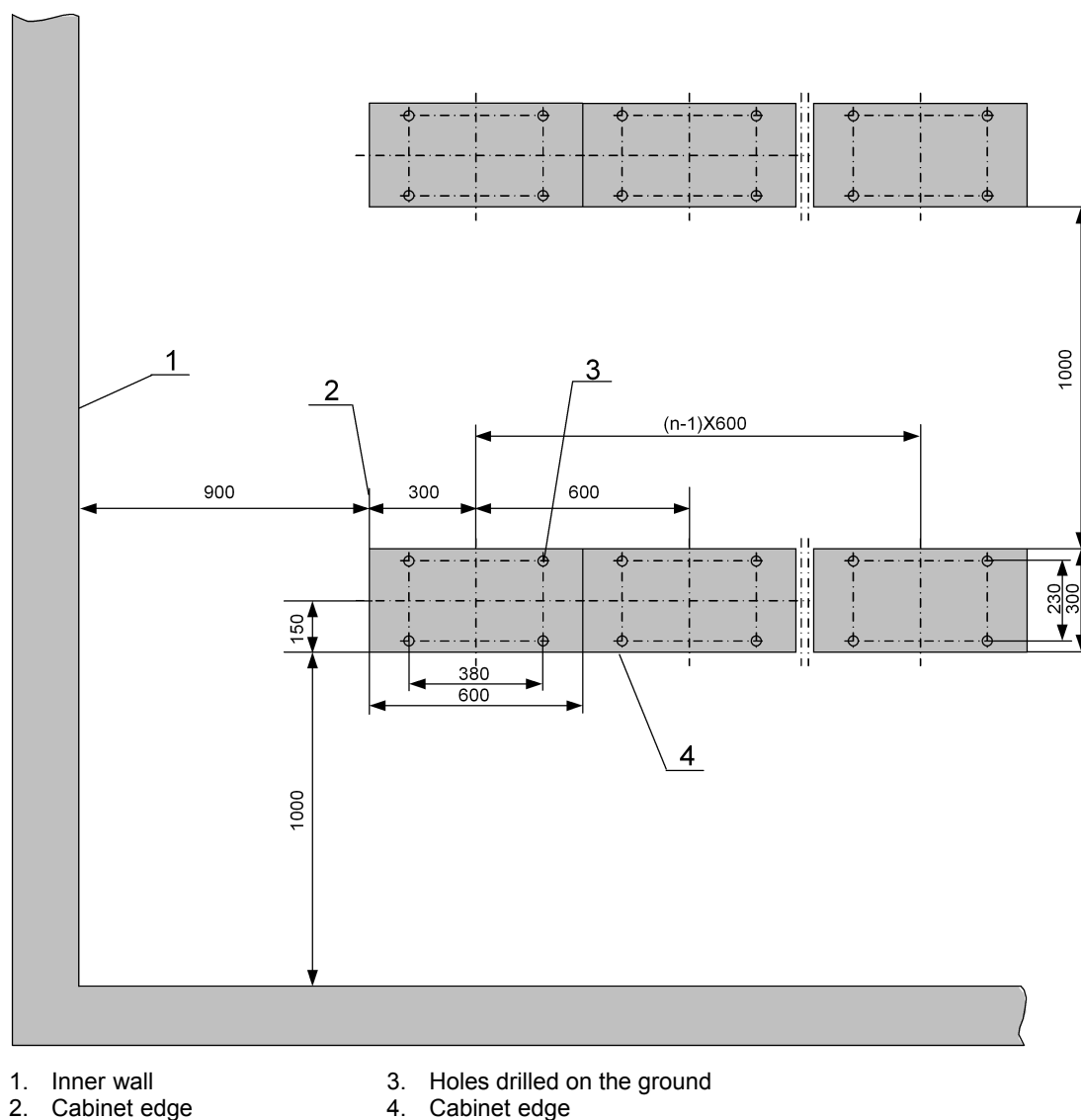
Figure 3-17 Fixing Two Cabinets In Parallel



- | | |
|-----------------|--------------------------------|
| 1. Inner wall | 3. Holes drilled on the ground |
| 2. Cabinet edge | 4. Cabinet edge |

Fixing Multiple Cabinets

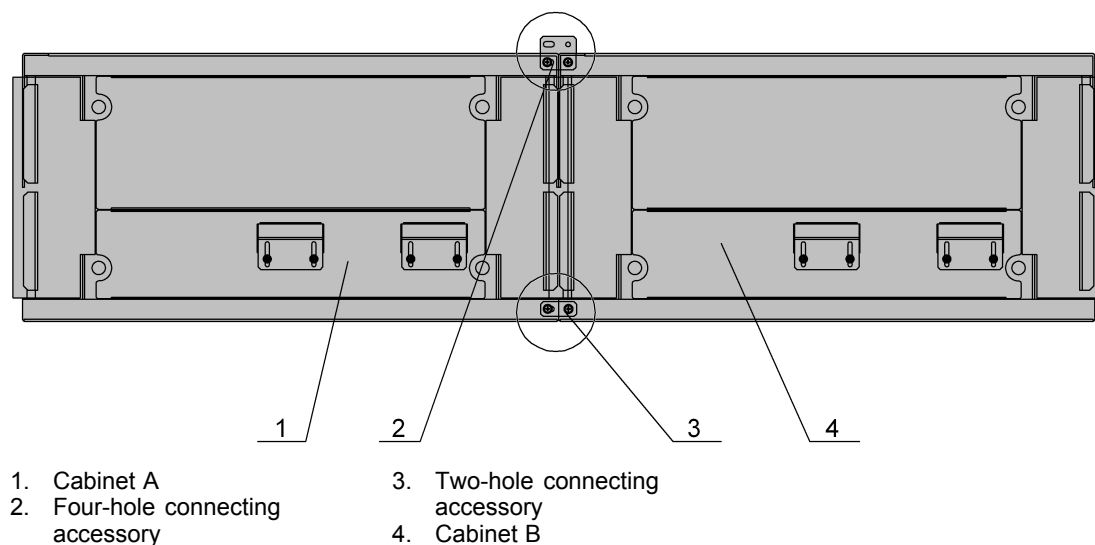
Fix multiple cabinets in parallel, as shown in [Figure 3-18](#).

Figure 3-18 Fixing Multiple Cabinets In Parallel

- 1. Inner wall
- 2. Cabinet edge
- 3. Holes drilled on the ground
- 4. Cabinet edge

3.4.3 Adjacent Cabinets Connection

When two or more than two cabinets are placed one by one, it is necessary to connect them on the top, as shown in [Figure 3-19](#).

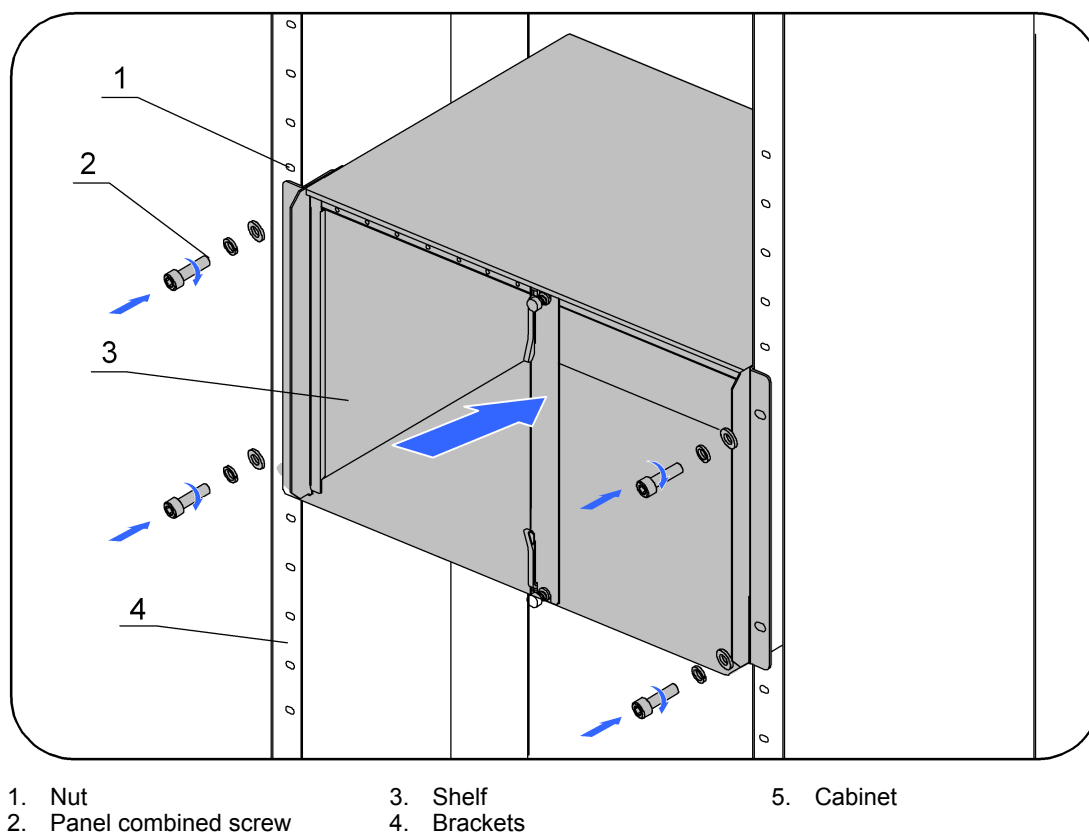
Figure 3-19 Connecting Adjacent Cabinets**Note:**

Four-hole connecting accessories are used for connecting cabinets back to back.

3.5 Shelf Installation

If the shelf is packed together with the cabinet, it is unnecessary to install the shelf. It is only required to check the screws and connecting cable. Make sure that the screws and connecting cables are fixed.

If the shelf is packed separately, it is necessary to install the shelf. [Figure 3-20](#) shows how to install a shelf in a cabinet.

Figure 3-20 Installing Shelf

3.6 Plug-in Box Installation

Plug-in box refers to the components such as the fan box, wiring box, and blank panel.

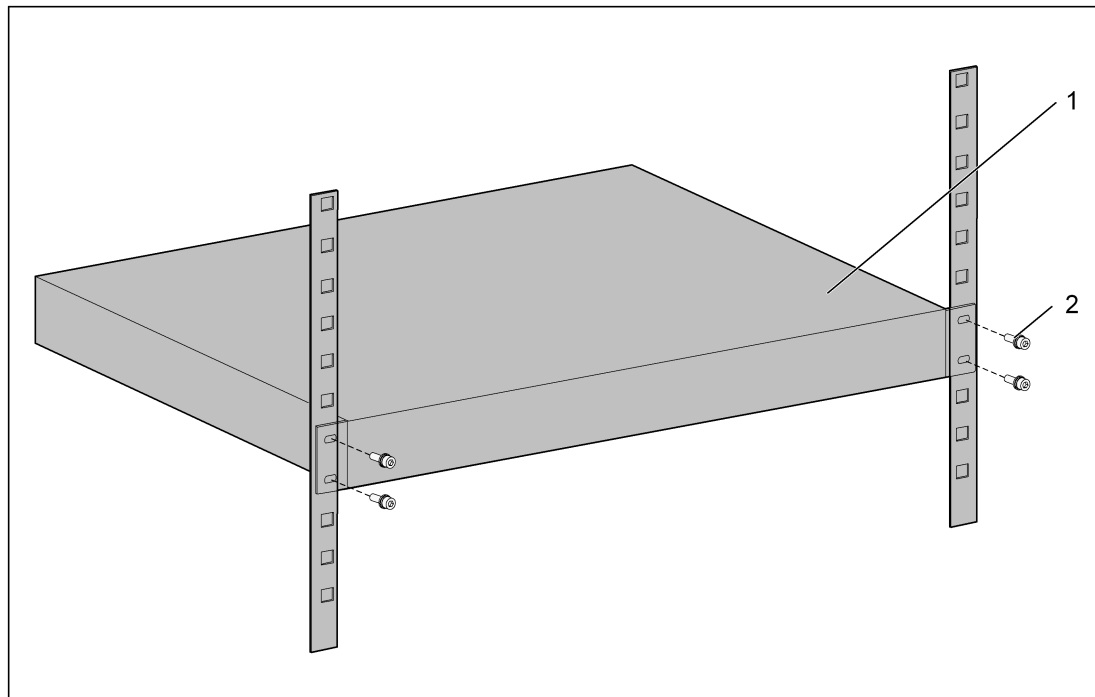
If plug-in boxes are packed together with the cabinet, it is unnecessary to install them. It is only required to check the screws and connecting cables. Make sure that the screws and connecting cables are fixed.

If plug-in boxes are packed separately, it is necessary to install them.

There are two position columns at both sides of the cabinet. To install a plug-in box in a cabinet, firstly, place the plug-in box on the brackets. Next, push it into the cabinet. Then, fasten M5 screws.

Figure 3-21 shows how to install a plug-in box in a cabinet.

Figure 3-21 Installing Plug-in Box



1. Fan box

2. M5 screw

3.7 Card Installation

This topic describes how to install a card.

Preparations

Perform the following steps to prepare for card installation:

1. Clean cards and shelves inside.
2. Take the card out of the antistatic bag.
3. Verify the card name according to the packing list.
4. Make sure that the card is not damaged and its components are complete.
5. Verify the software version of the card.
6. Make sure that conductive foam gaskets are installed properly.

Installing

Perform the following steps to install a card:

1. Wear the antistatic wrist strap.
2. Insert the card into the correct slot.
3. Press clasps on the card and push it in until it is latched, which means the card is in place.

On the back of each shelf, there are floating grounding spring plates, which enable all cards to be grounded.

Installing Dummy Panel

Install dummy panels in empty slots. The spring plate at the back of the dummy panel is used to adjust the tightness.

Chapter 4

Power Supply Installation

Table of Contents

Power and Grounding Cables.....	4-1
Installation Flowchart.....	4-1
Power and Grounding Cable Structure	4-2
Power and Grounding Cable Installation.....	4-4

4.1 Power and Grounding Cables

Power cables include the following:

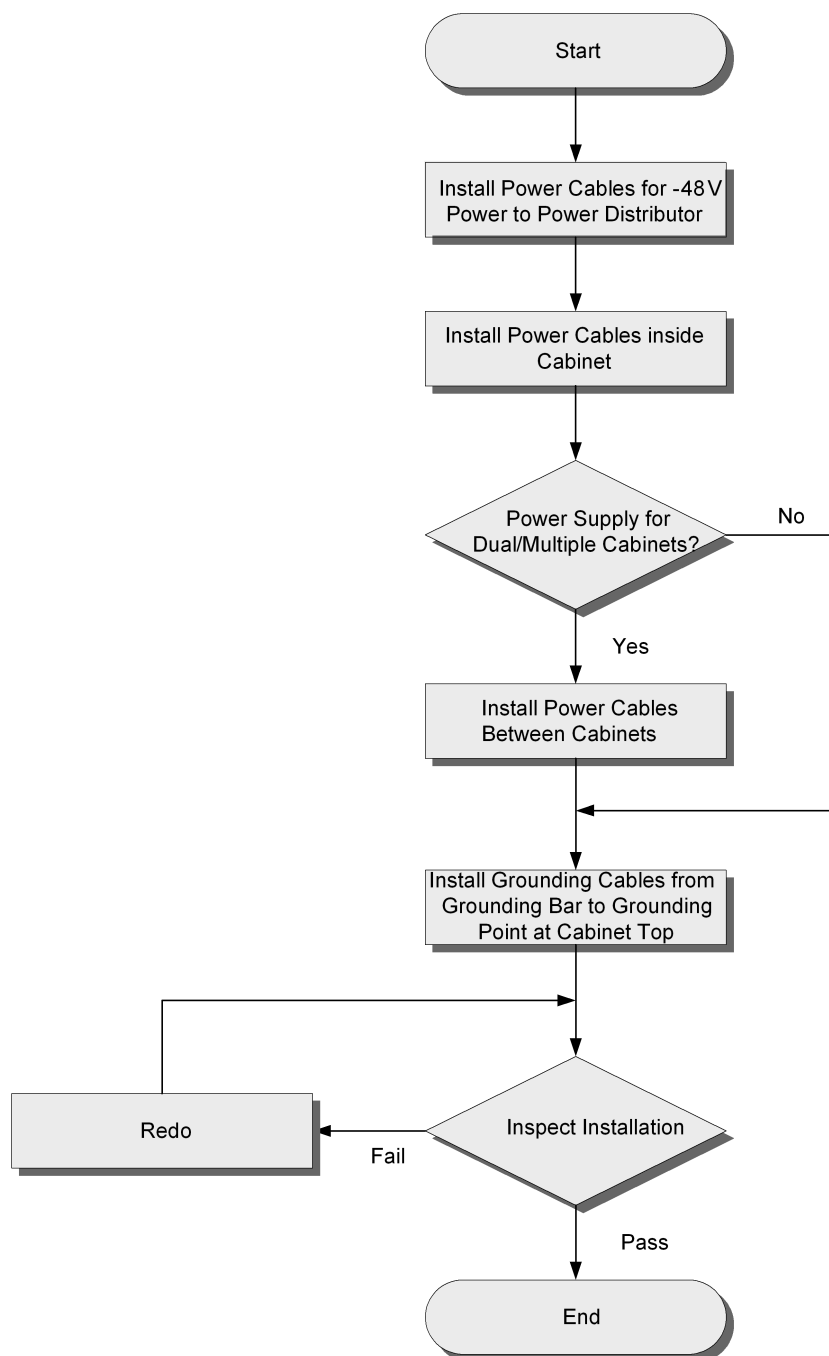
- -48 V input power cables
- Power cables inside the cabinet

Grounding cables are cabinet protection cables.

4.2 Installation Flowchart

Figure 4-1 shows the power and grounding cables installation flowchart.

Figure 4-1 Power and Grounding Cables Installation Flowchart



4.3 Power and Grounding Cable Structure

-48 V Input Power Cable

Figure 4-2 shows the -48 V input power cables structure.

Figure 4-2 -48 V Power Cable Structure



End A connects to the power distributor and End B connects to the power distribution cabinet in the equipment room. The blue power cable 1 is a -48 V power cable and the black power cable 2 is a -48 V grounding power cable.

Internal Power Cable

Figure 4-3 shows the internal power cable structure.

Figure 4-3 Internal Power Cable Structure

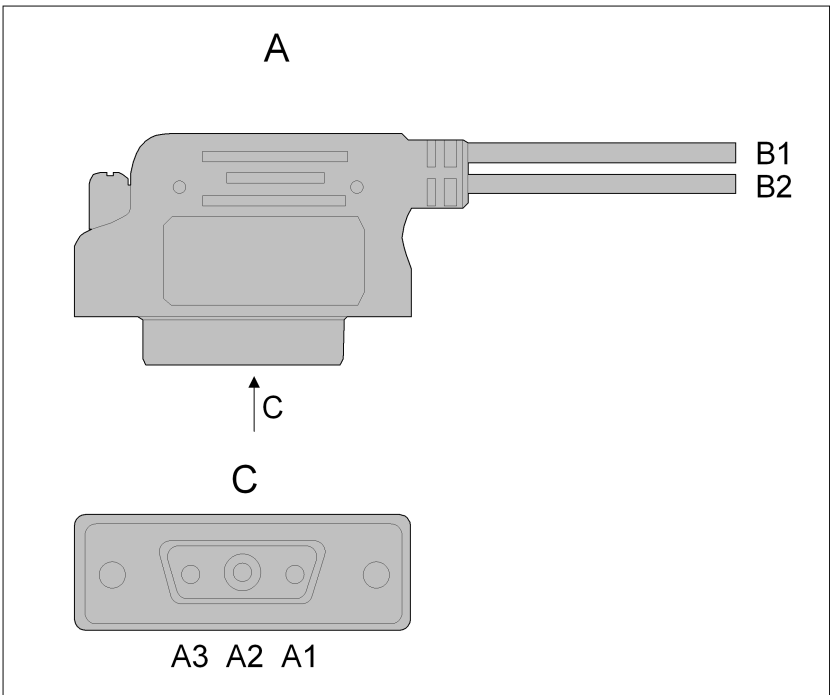


Table 4-1 lists the internal power cable connections.

Table 4-1 Internal Power Cable Pin Connections

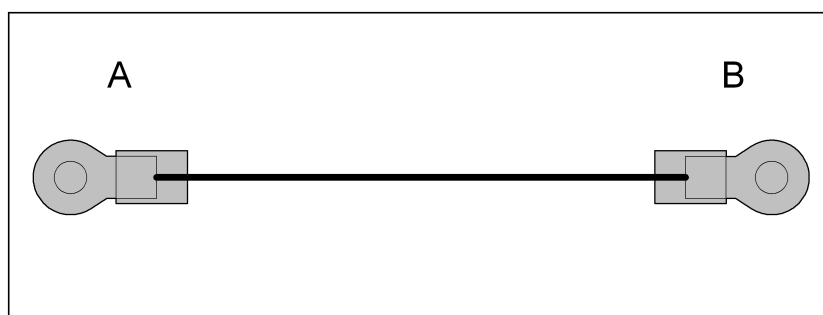
End A (Connects to Power Card)	Power Cable Color Spectrum/Heat Shrinkage Bush	Signal Description	End B (Connects to Power Distributor)
A1	Black/Blue Bush	-48 V	B1

End A (Connects to Power Card)	Power Cable Color Spectrum/Heat Shrinkage Bush	Signal Description	End B (Connects to Power Distributor)
A3	Black/Without Bush	-48 VRTN	B2

Protection Grounding Cable

Figure 4-4 shows the protection grounding cable structure.

Figure 4-4 Protection Grounding Cable Structure



Connect one end of the protection grounding cable to the grounding bar in the equipment room and the other end to the grounding point at the top of the cabinet. The protection grounding cable is yellow/green.

4.4 Power and Grounding Cable Installation

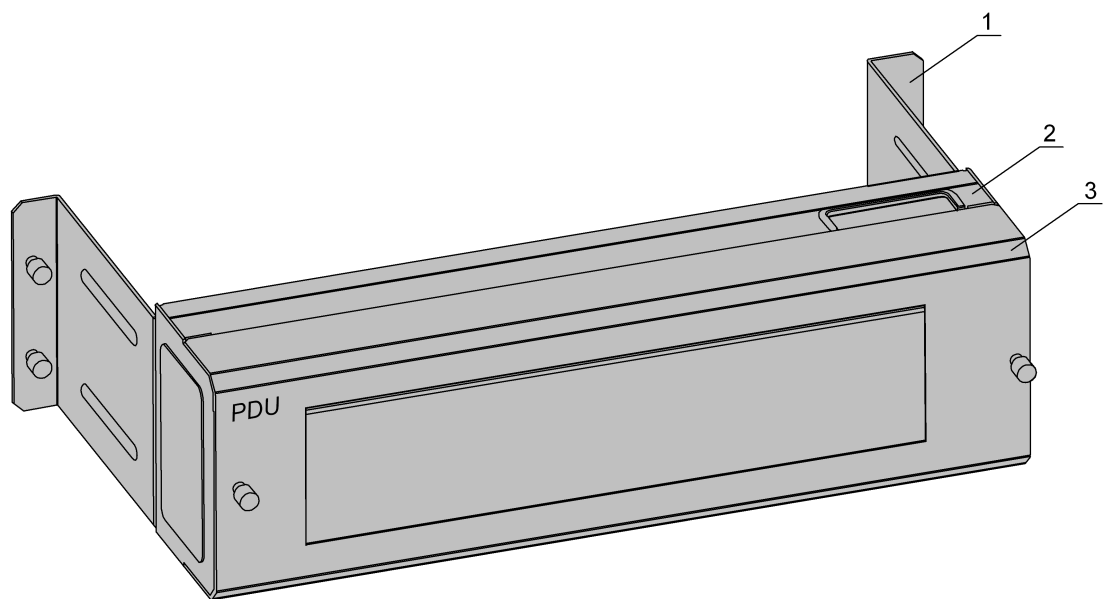
This topic includes the following:

- 19-Inch Power Distributor
- -48 V Input Power Cable Installation
- Internal Power Cable Installation
- Grounding Cable Installation

4.4.1 19-Inch Power Distributor

Dimensions

The dimensions of the 19-inch power distributor are 131 mm × 482.6 mm × 240 mm (Height × Width × Depth), as shown in Figure 4-5.

Figure 4-5 19-Inch Power Distributor Dimensions

1. Installation bracket

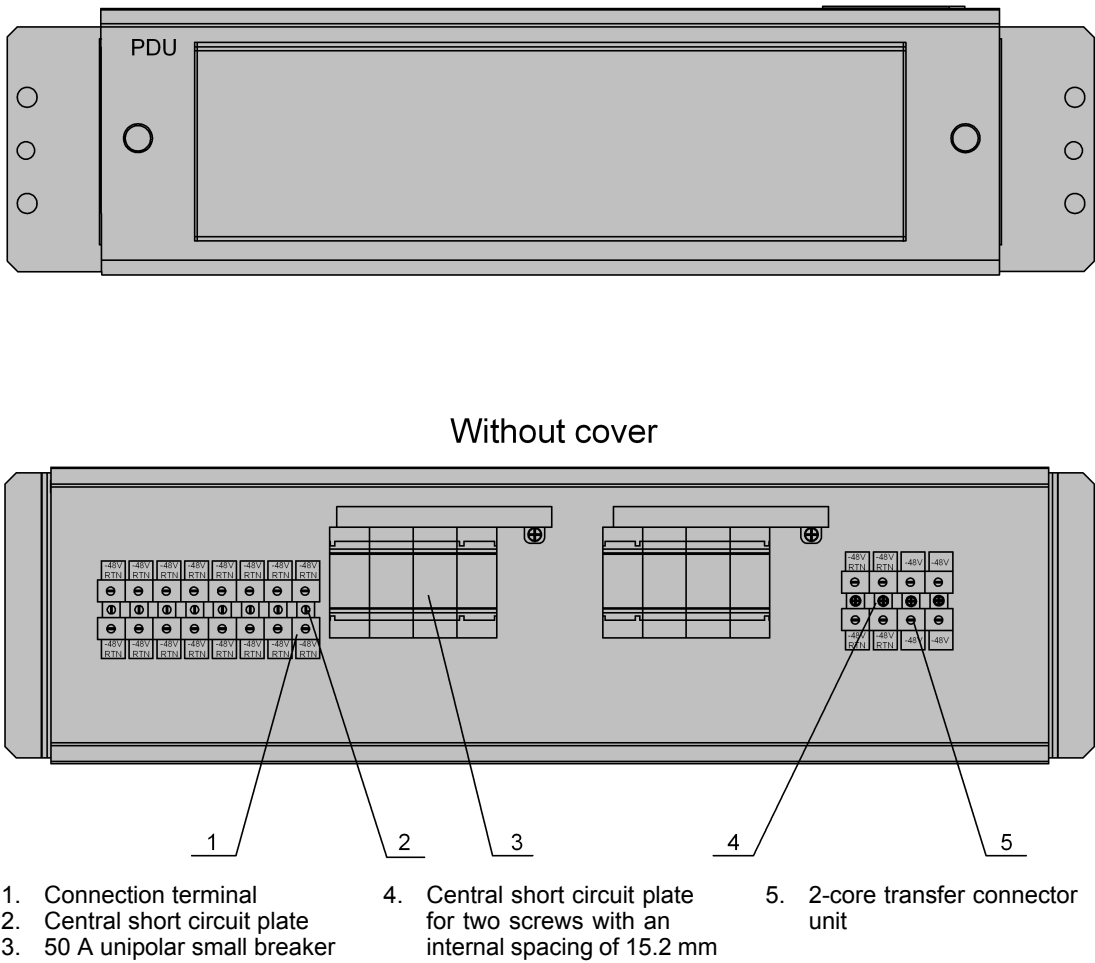
2. Box bottom

3. Box cover

Internal Structure

Figure 4-6 shows the power distributor internal structure.

Figure 4-6 Power Distributor Internal Structure

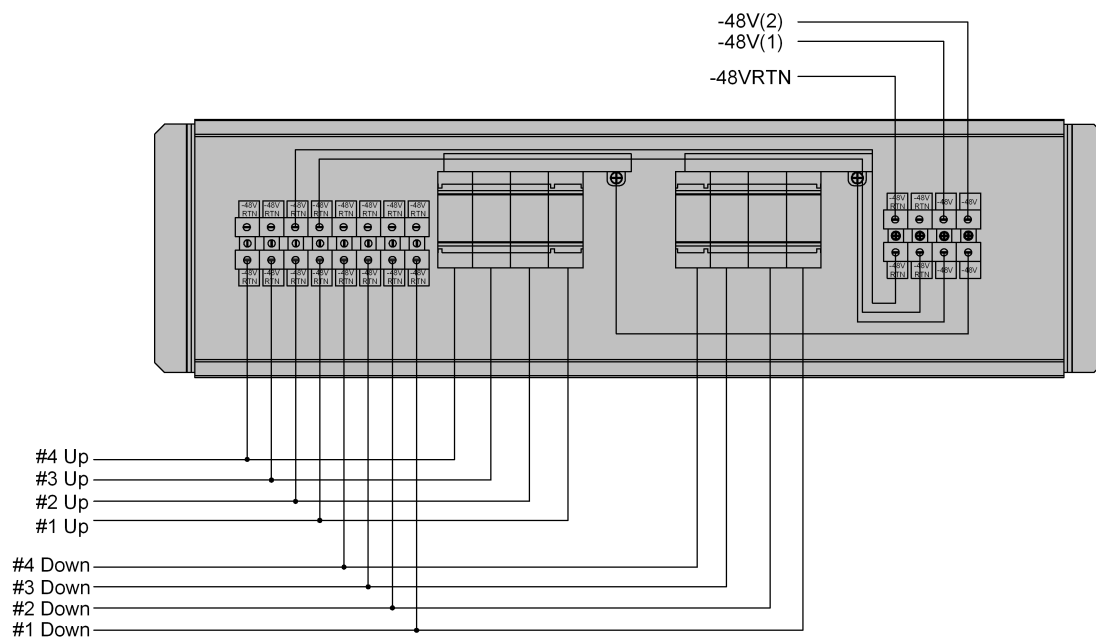


4.4.2 -48 V Input Power Cable Installation

Power cables includes -48 V input power cables and the power cables inside the cabinet. The power distributor leads in the -48 V power, and then distributes it to different shelves.

Figure 4-7 shows the cable connections for the power distributor.

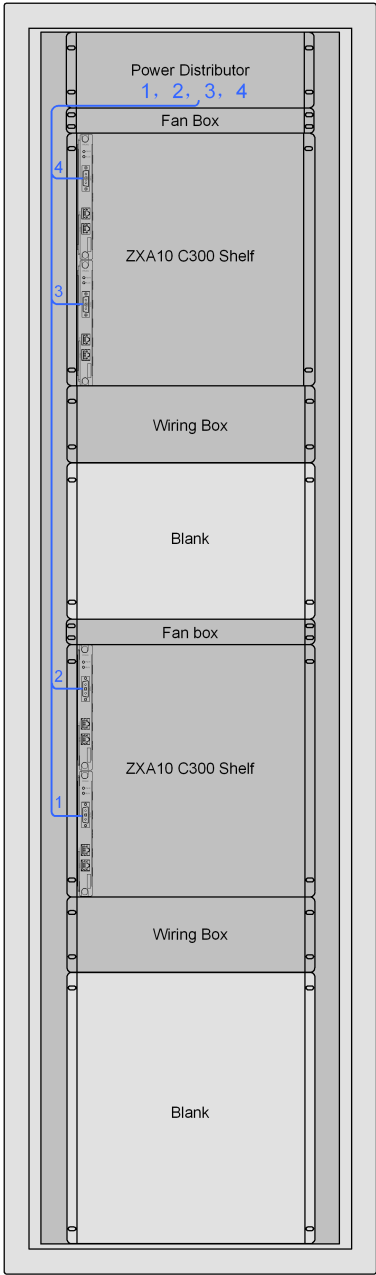
Figure 4-7 Power Distributor Cable Connections



4.4.3 Internal Power Cable Installation

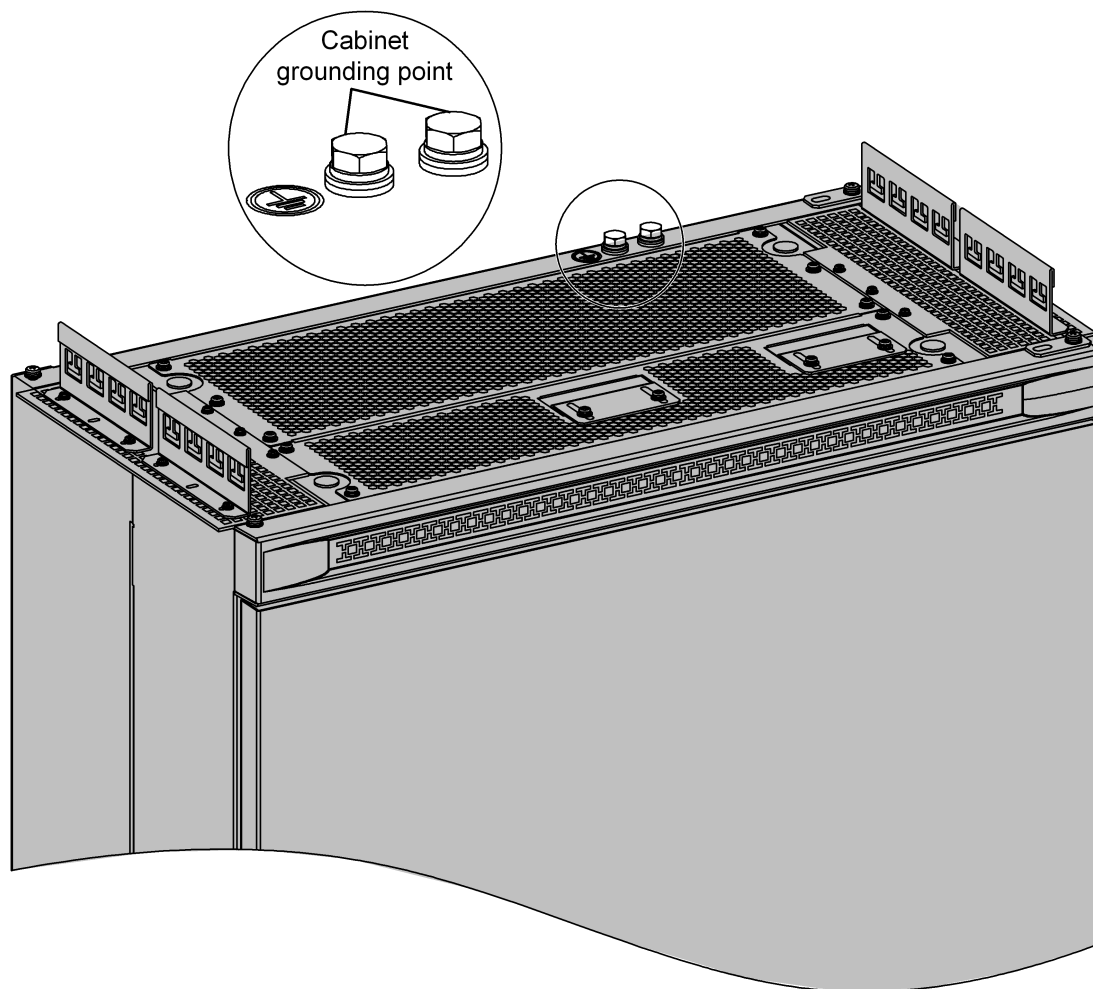
Figure 4-8 shows the power cable layout inside the cabinet.

Figure 4-8 Internal Power Cable Layout



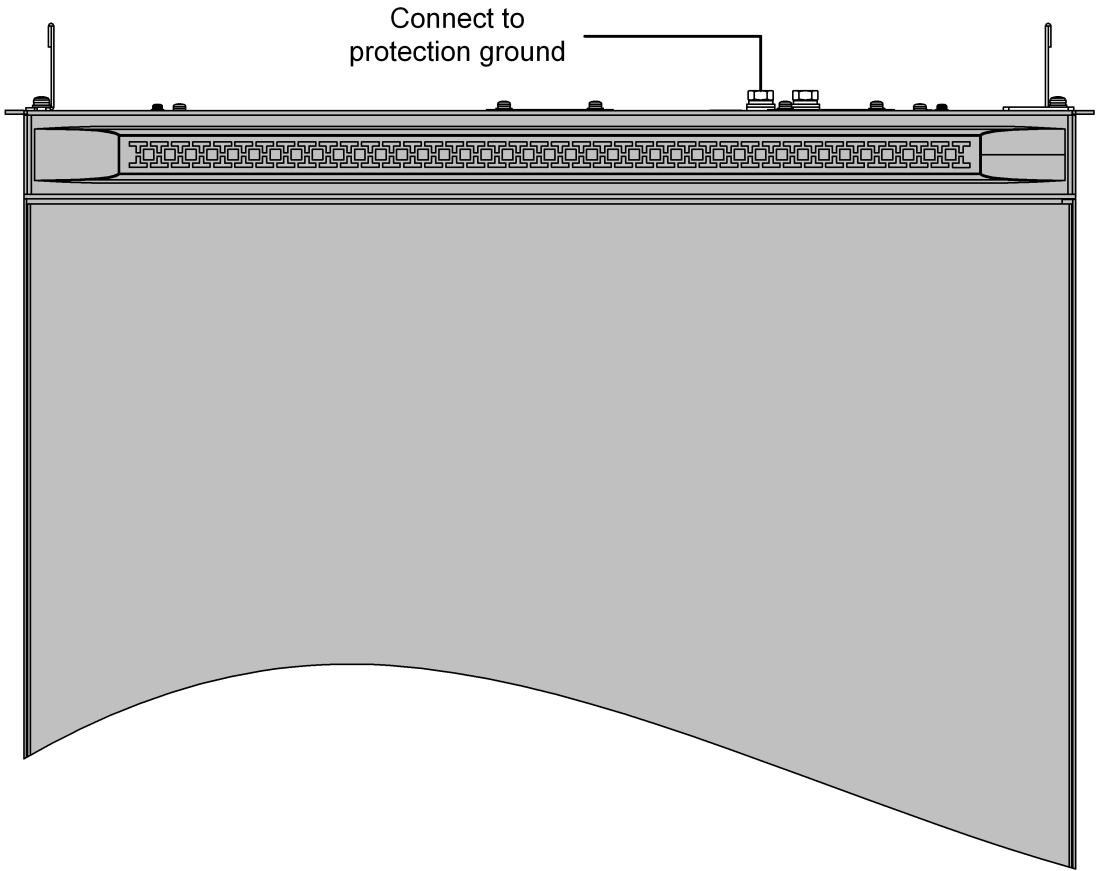
4.4.4 Grounding Cable Installation

The cabinet grounding point is at the top of the cabinet, as shown in [Figure 4-9](#).

Figure 4-9 Cabinet Grounding Point

Connect the protection ground from the grounding busbar in the equipment room to the grounding point at the top of the cabinet, as shown in [Figure 4-10](#).

Figure 4-10 Protection Ground Connection



Chapter 5

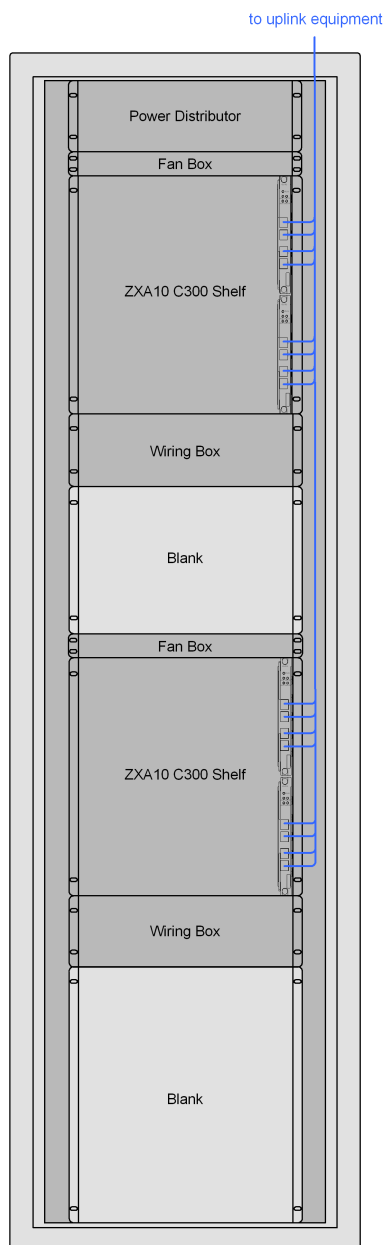
Cable Connection

Table of Contents

Overall Cabling.....	5-1
Cable Layout.....	5-2
E1/T1 Cable	5-6
Monitoring Cables	5-10
Serial Port Cable	5-16
Network Cable.....	5-18
Optical Fiber.....	5-19

5.1 Overall Cabling

Figure 5-1 shows the uplink cabling in the cabinet.

Figure 5-1 Uplink Cabling

5.2 Cable Layout

This topic includes the following:

- Cabling Principles
- Cable Bundling

5.2.1 Cabling Principles

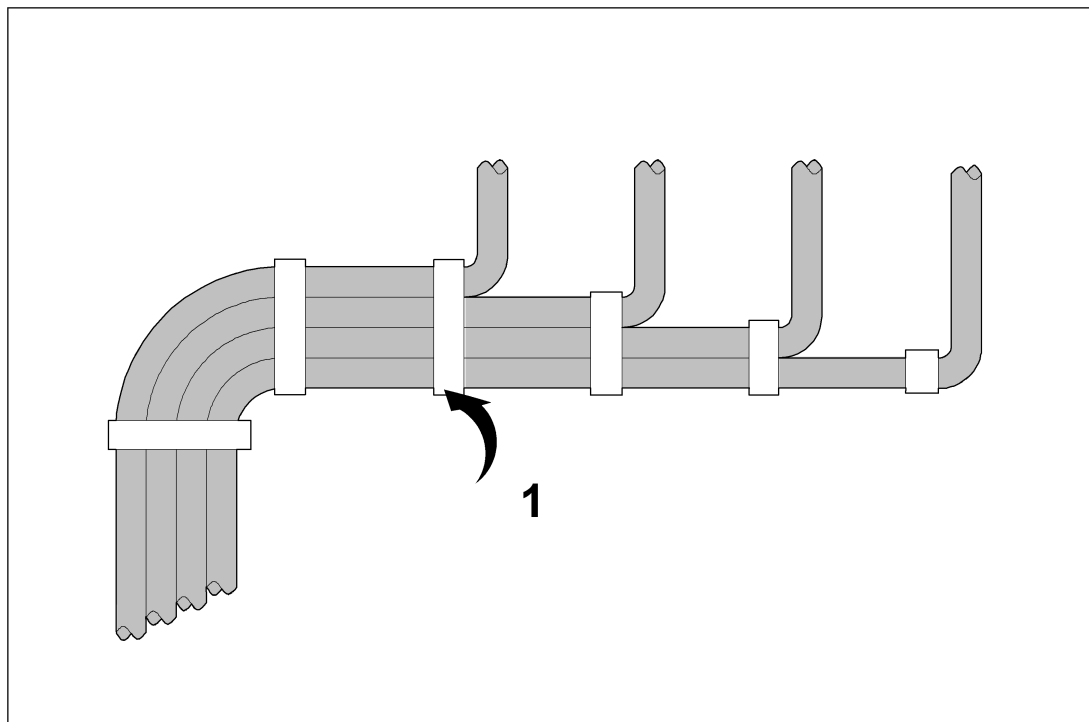
The basic principles for cabling are as follows:

- Confirm cable specifications, quantity, and type according to the construction design and contract.
- Check quality certificates, such as ex-factory record and quality warranty.
- Route and position cables according to the construction design.
- Decide the cable length according to the real situation.
- Use the one-piece cable without any joints.
- Arrange the cables neatly.
- Bend the cables smoothly and evenly. Curvature radius of cable bends should be at least 60 mm for cables whose outer diameter is less than 12 mm. For cables having diameter more than 12 mm, the radius of curvature should be 10 times of the diameter.
- Lay subscriber cables separately from power cables.
- Keep cables straight within troughs. Ensure that cables are not blocking or interrupting other cable inlets or outlets.
- If fiber, power, and grounding cables are laid in same trough, ensure that cables do not overlap each other. Roll up extra cables neatly and fix them to the middle of cabling rack.
- Carefully handle the tail fibers. Use protective plastic bushings and tail fibers should be bundled around the top or bottom of the cabinet.
- The radius of fibers bends should be at least 38 mm to 40 mm within troughs. Keep bending fibers straight within troughs.
- Label each cable.

5.2.2 Cable Bundling

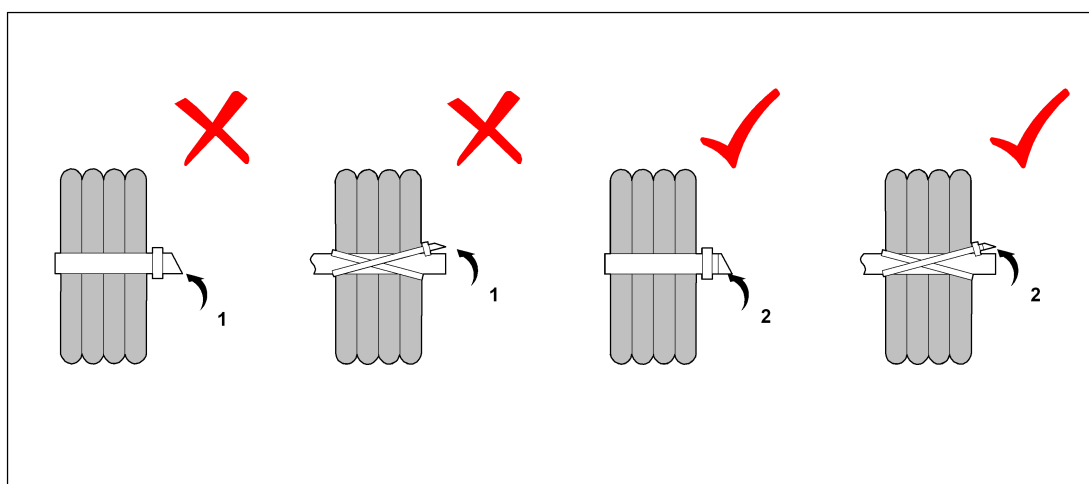
Cable Binding Technique

Bundle cables in sequence. Avoid bending the cables. [Figure 5-2](#) shows the cables bundling technique. Keep cables vertical to the ground. Bind cables with a tie at an equal distance.

Figure 5-2 Cables Bundling

Tie Cutting Technique

Cut the tie head smoothly. [Figure 5-3](#) shows the tie cutting technique. The tick mark shows a correct cutting while the cross mark shows an incorrect cutting.

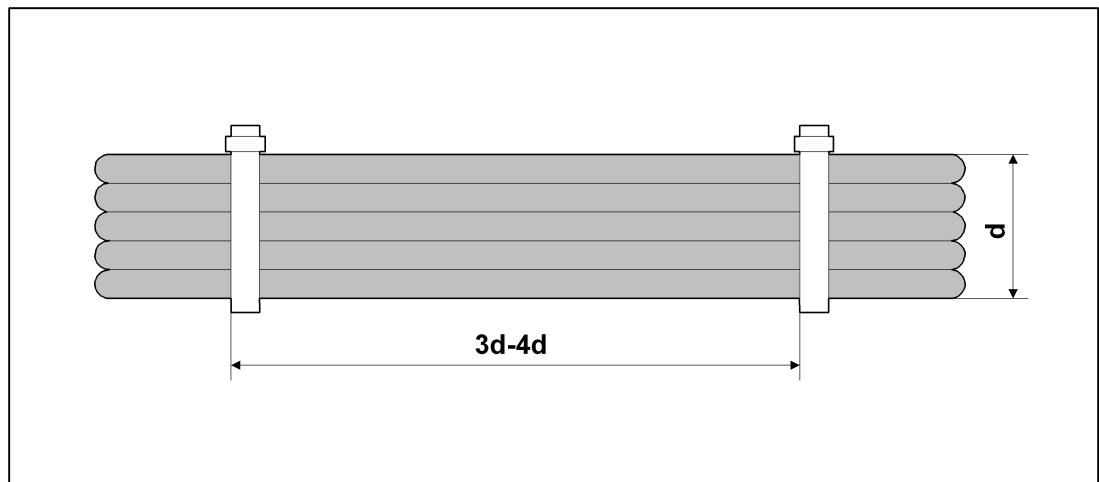
Figure 5-3 Tie Cutting

1. Sharp end
2. Cut smoothly

Tie Space

Keep a space between the adjacent ties. The space is two or three times of the cables bind diameter, as shown in [Figure 5-4](#).

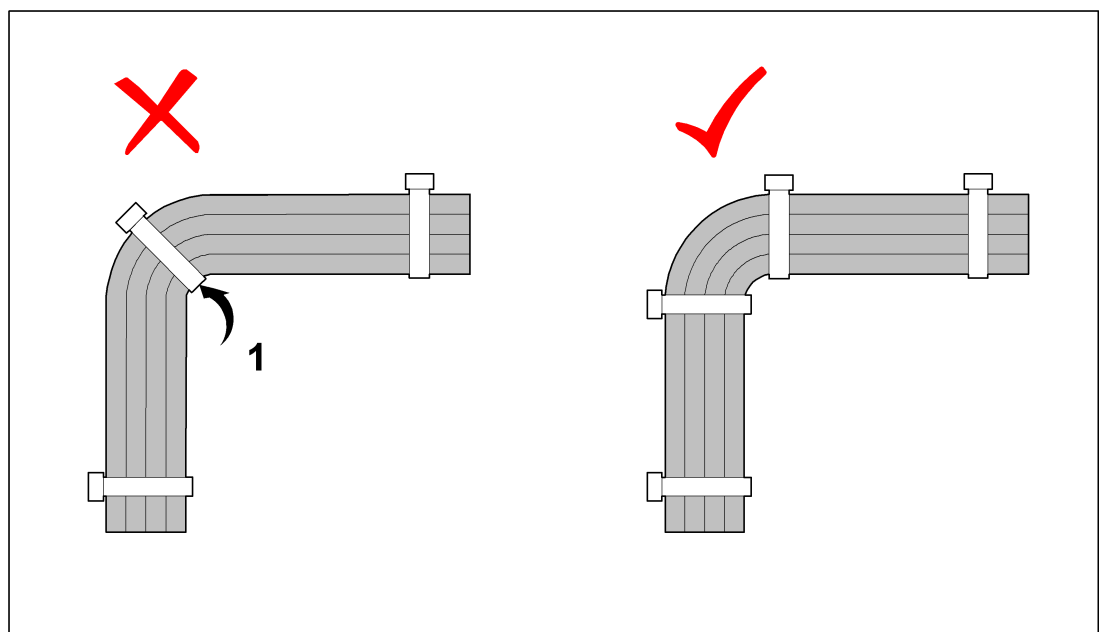
Figure 5-4 Tie Space



Tie Position

Keep ties at the correct positions, as shown in [Figure 5-5](#).

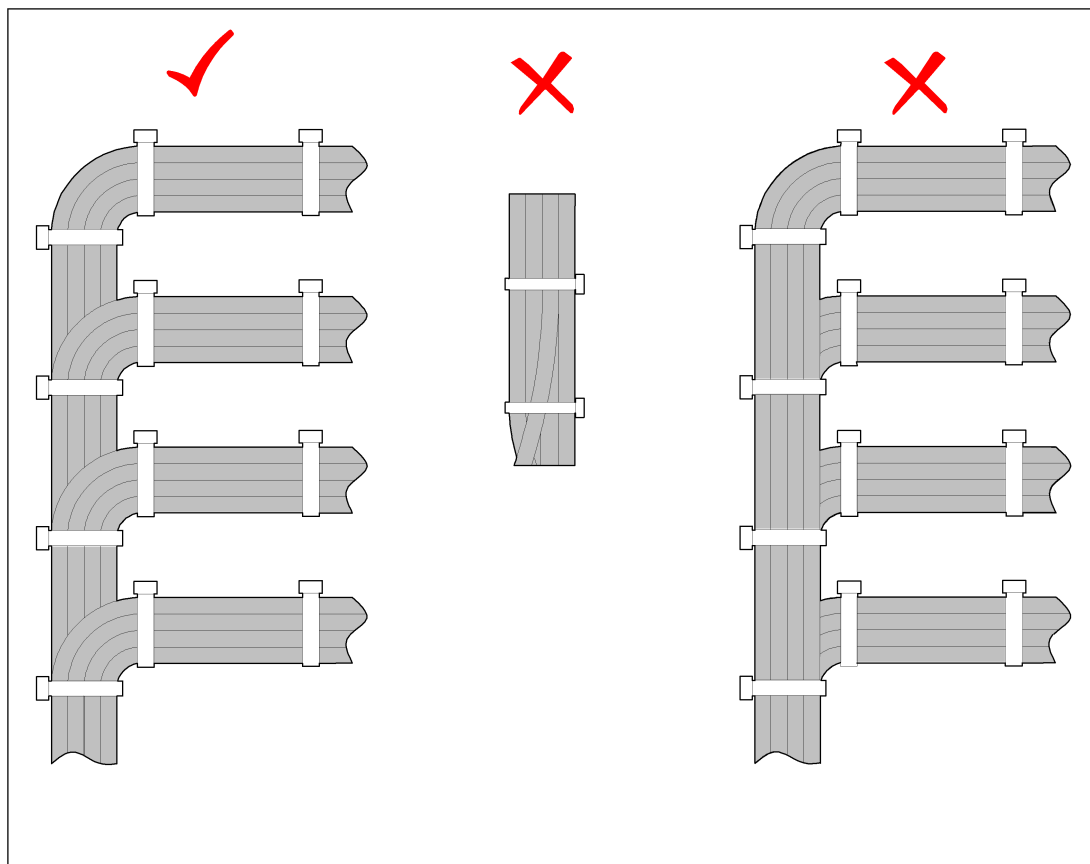
Figure 5-5 Cable Tie Position



1. Cable tie

[Figure 5-6](#) shows the cabling technique in cabinet.

Figure 5-6 Cabling Technique In Cabinet



5.3 E1/T1 Cable

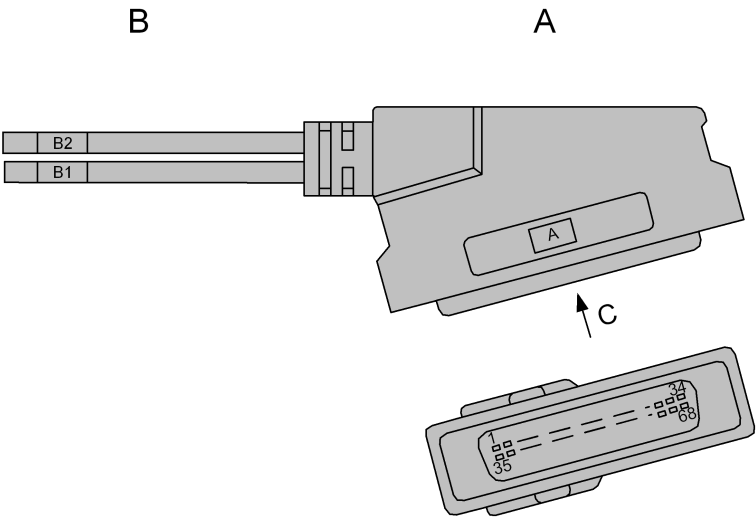
The E1 cable is used to connect the CTBB card to the [PSTN](#) network. The T1 cable is used to connect the CTTB card to the PSTN network.

E1 Cable

- E1 cable structure

[Figure 5-7](#) shows the E1 cable structure.

Figure 5-7 E1 Cable Structure



- E1 cable pin connections
- Table 5-1 lists E1 cable pin connections.

Table 5-1 E1 Cable Pin Connections

Pin No. at End A	Cable Color Spectrum B1		Pin No. at End A	Cable Color Spectrum B2
1	Blue (Red1)		17	Blue (Red1)
35	Blue (Black1)		51	Blue (Black1)
2	Pink (Red1)		18	Pink (Red1)
36	Pink (Black1)		52	Pink (Black1)
3	Green (Red1)		19	Green (Red1)
37	Green (Black1)		53	Green (Black1)
4	Orange (Red1)		20	Orange (Red1)
38	Orange (Black1)		54	Orange (Black1)
5	Grey (Red1)		21	Grey (Red1)
39	Grey (Black1)		55	Grey (Black1)
6	Blue (Red2)		22	Blue (Red2)
40	Blue (Black2)		56	Blue (Black2)
7	Pink (Red2)		23	Pink (Red2)
41	Pink (Black2)		57	Pink (Black2)
8	Green (Red2)		24	Green (Red2)
42	Green (Black2)		58	Green (Black2)
9	Orange (Red2)		25	Orange (Red2)

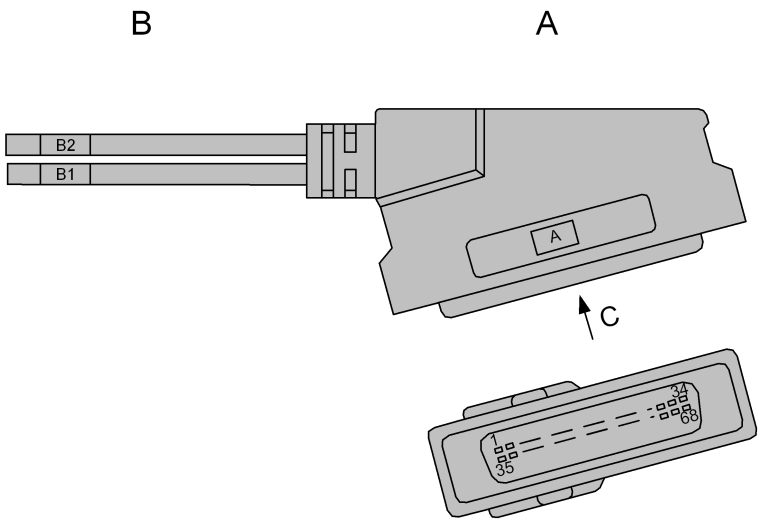
Pin No. at End A	Cable Color Spectrum B1		Pin No. at End A	Cable Color Spectrum B2
43	Orange (Black2)		59	Orange (Black2)
10	Grey (Red2)		26	Grey (Red2)
44	Grey (Black2)		60	Grey (Black2)
11	Blue (Red3)		27	Blue (Red3)
45	Blue (Black3)		61	Blue (Black3)
12	Pink (Red3)		28	Pink (Red3)
46	Pink (Black3)		62	Pink(Black3)
13	Green (Red3)		29	Green (Red3)
47	Green (Black3)		63	Green (Black3)
14	Orange (Red3)		30	Orange (Red3)
48	Orange (Black3)		64	Orange (Black3)
15	Grey (Red3)		31	Grey (Red3)
49	Grey (Black3)		65	Grey (Black3)
16	Blue (Red4)		32	Blue (Red4)
50	Blue (Black4)		66	Blue (Black4)

T1 Cable

- T1 cable structure

Figure 5-8 shows the T1 cable structure.

Figure 5-8 T1 Cable Structure



- T1 cable pin connections

Table 5-2 lists T1 cable pin connections.

Table 5-2 T1 Cable Pin Connections

Pin No. at End A	Cable Color Spectrum B1		Pin No. at End A	Cable Color Spectrum B2
1	Blue (Red1)		17	Blue (Red1)
35	Blue (Black1)		51	Blue (Black1)
2	Pink (Red1)		18	Pink (Red1)
36	Pink (Black1)		52	Pink (Black1)
3	Green (Red1)		19	Green (Red1)
37	Green (Black1)		53	Green (Black1)
4	Orange (Red1)		20	Orange (Red1)
38	Orange (Black1)		54	Orange (Black1)
5	Grey (Red1)		21	Grey (Red1)
39	Grey (Black1)		55	Grey (Black1)
6	Blue (Red2)		22	Blue (Red2)
40	Blue (Black2)		56	Blue (Black2)
7	Pink (Red2)		23	Pink (Red2)
41	Pink (Black2)		57	Pink (Black2)
8	Green (Red2)		24	Green (Red2)
42	Green (Black2)		58	Green (Black2)
9	Orange (Red2)		25	Orange (Red2)
43	Orange (Black2)		59	Orange (Black2)
10	Grey (Red2)		26	Grey (Red2)
44	Grey (Black2)		60	Grey (Black2)
11	Blue (Red3)		27	Blue (Red3)
45	Blue (Black3)		61	Blue (Black3)
12	Pink (Red3)		28	Pink (Red3)
46	Pink (Black3)		62	Pink (Black3)
13	Green (Red3)		29	Green (Red3)
47	Green (Black3)		63	Green (Black3)
14	Orange (Red3)		30	Orange (Red3)
48	Orange (Black3)		64	Orange (Black3)
15	Grey (Red3)		31	Grey (Red3)
49	Grey (Black3)		65	Grey (Black3)
16	Blue (Red4)		32	Blue (Red4)

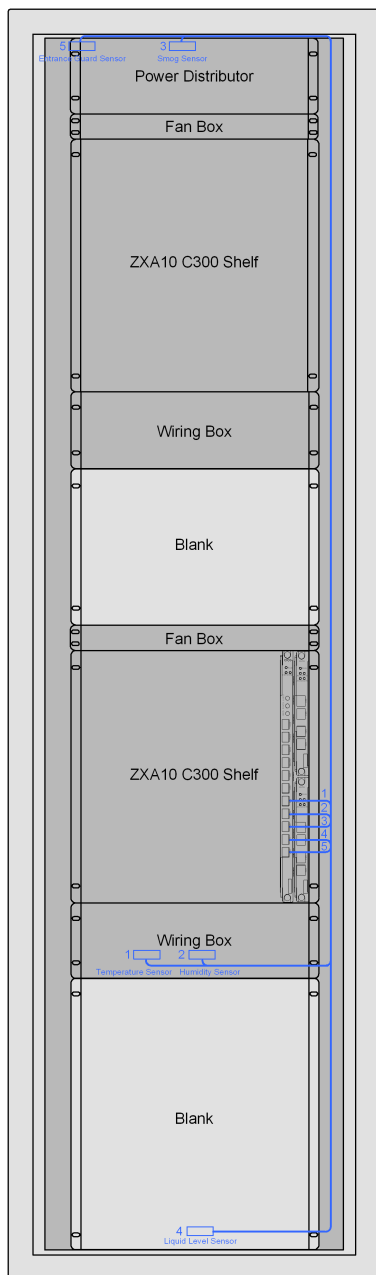
Pin No. at End A	Cable Color Spectrum B1		Pin No. at End A	Cable Color Spectrum B2
50	Blue (Black4)		66	Blue (Black4)

5.4 Monitoring Cables

Monitoring cables include the following:

- Entrance control monitoring cable
- Smog monitoring cable
- Flood monitoring cable
- Temperature monitoring cable
- Humidity monitoring cable

Figure 5-9 shows the monitoring cable routing in the cabinet.

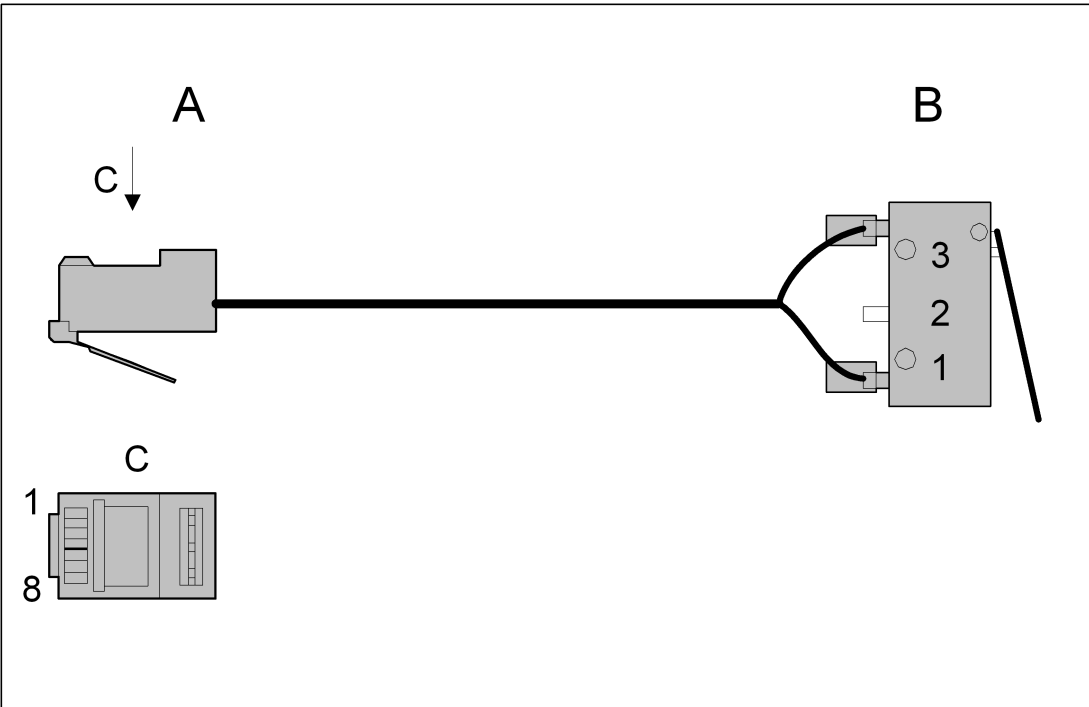
Figure 5-9 Monitoring Cable Routing

5.4.1 Entrance Control Monitoring Cable

Cable Structure

Figure 5-10 shows the entrance control monitoring cable structure.

Figure 5-10 Entrance Control Monitoring Cable Structure



Pin Connection

Table 5-3 lists the door control monitoring cable pin connections.

Table 5-3 Entrance Control Monitoring Cable Pin Connections

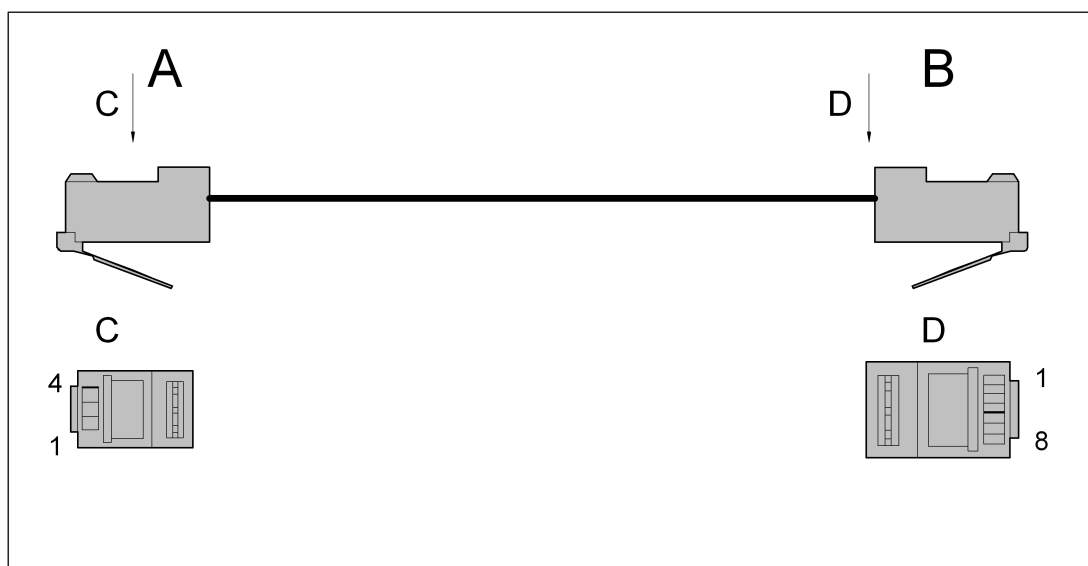
Pin No. at End A	Pin No. at End B	Signal Description
1	1	Switch A
4	2	Switch B

5.4.2 Smog Monitoring Cable

Cable Structure

Figure 5-11 shows the smog monitoring cable structure.

Figure 5-11 Smog Monitoring Cable Structure



Pin Connection

Table 5-4 lists smog monitoring cable pin connections.

Table 5-4 Smog Monitoring Cable Pin Connections

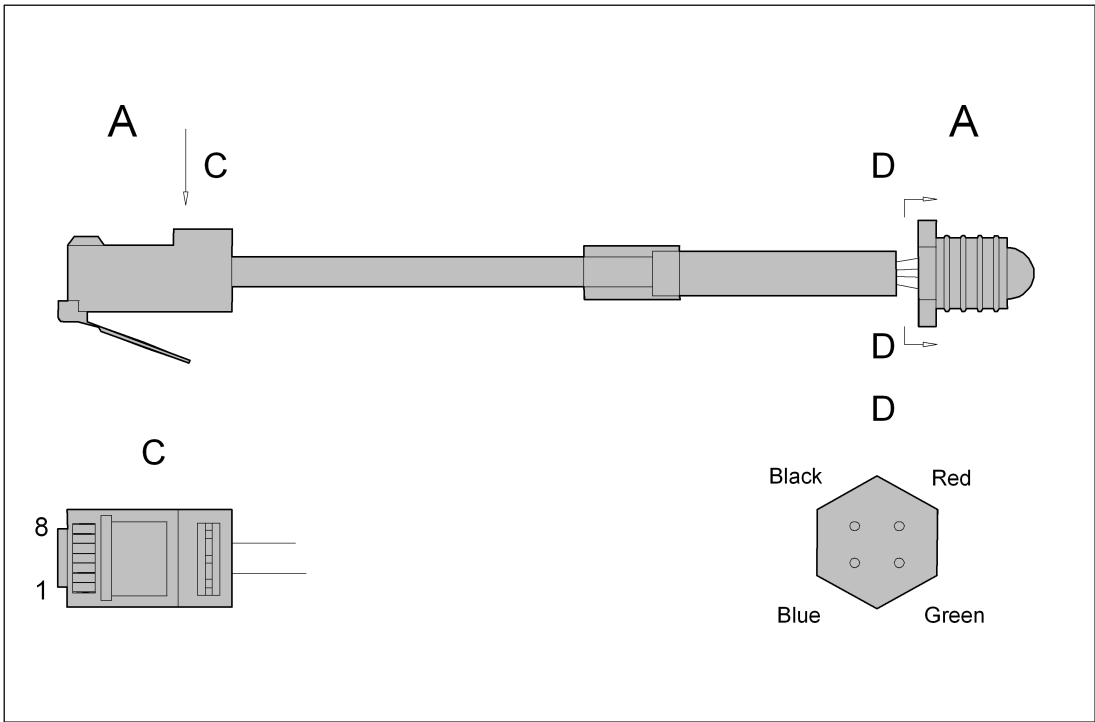
Pin No. at End A	Pin No. at End B	Signal Description
8	4	+12 V
4	2	Smog signal
1	1	GND

5.4.3 Flood Monitoring Cable

Cable Structure

Figure 5-12 the shows flood monitoring cable structure.

Figure 5-12 Flood Monitoring Cable Structure



Pin Connection

Table 5-5 lists flood monitoring cable pin connections.

Table 5-5 Flood Monitoring Cable Pin Connections

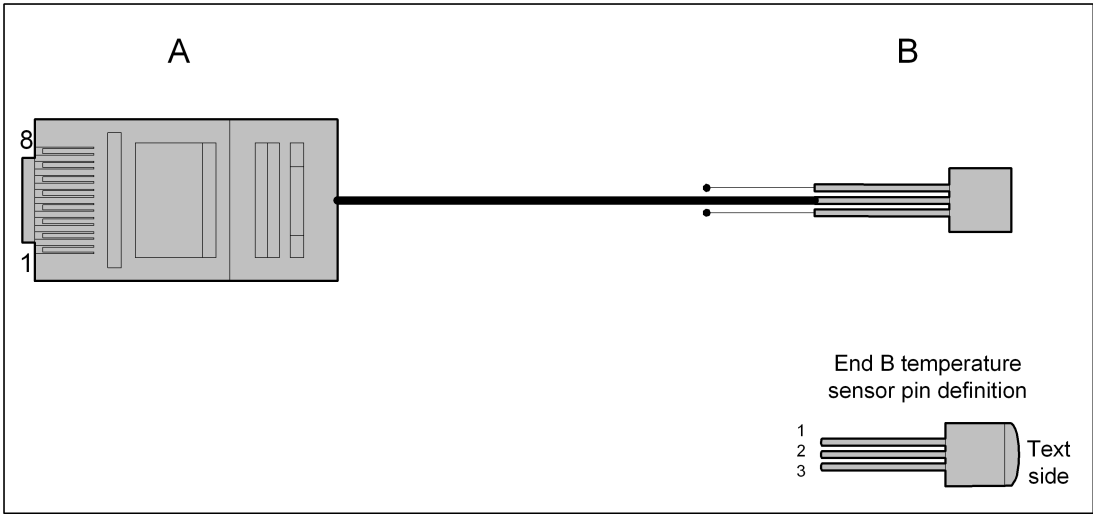
Pin No. at End A	End B	Signal Description
1	Blue cable	GND
4	Green cable	Flood signal
5	Black cable	Luminescent tube power
7	Red cable	+5 V

5.4.4 Temperature Monitoring Cable

Cable Structure

Figure 5-13 shows the temperature monitoring cable structure.

Figure 5-13 Temperature Monitoring Cable Structure



Pin Connection

Table 5-6 lists temperature monitoring cable pin connections.

Table 5-6 Temperature Monitoring Cable Pin Connections

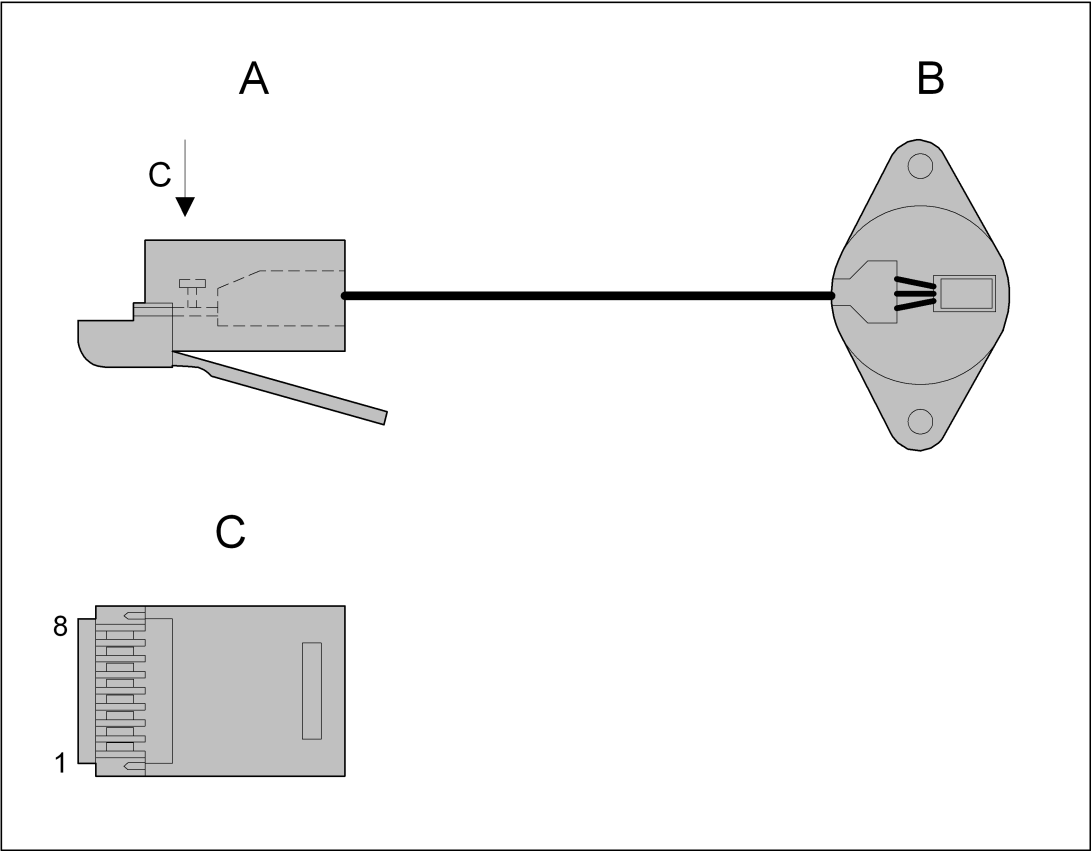
Pin No. at End A	Pin No. at End B	Signal Description
1	3	+5 V
4	2	Temperature Signal
7	1	GND

5.4.5 Humidity Monitoring Cable

Cable Structure

Figure 5-14 shows the humidity monitoring cable structure.

Figure 5-14 Humidity Monitoring Cable Structure



Pin Connection

Table 5-7 lists humidity monitoring cable pin connections.

Table 5-7 Humidity Monitoring Cable Pin Connections

Pin No. at End A	Pin No. at End B	Signal Description
7	3	+5 V
4	2	Humidity signal
1	1	GND

5.5 Serial Port Cable

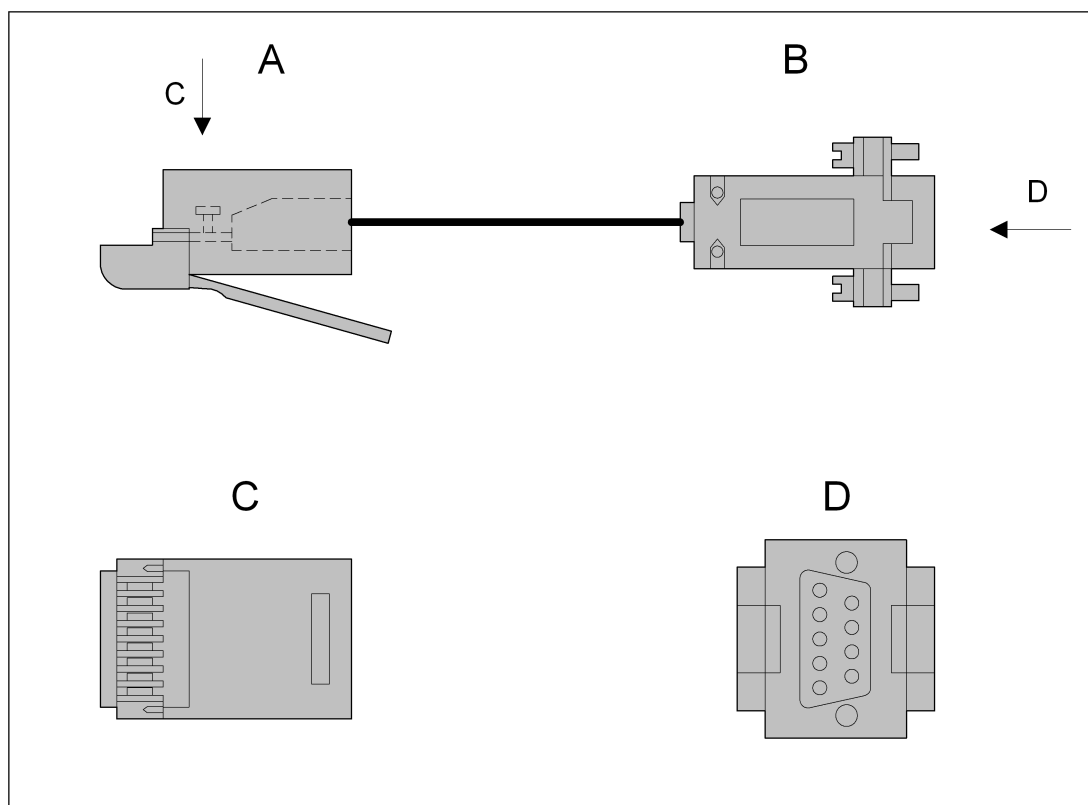
Overview

The serial port cable connects the serial port of the local maintenance computer to the ZXA10 C300 equipment.

Serial Port Cable Structure

Figure 5-15 shows the serial port cable structure. End A is an RJ45 connector and End B is a DB9 serial port connector.

Figure 5-15 Serial Port Cable Structure



Serial Port Cable Connections

Table 5-8 lists the serial port cable connections.

Table 5-8 Serial Port Cable Connections

End A Connecting Console Port on GCSD	End B Connecting Local Serial Port
3	3
4	5
5	5
6	2
Note: Other pins are null.	

5.6 Network Cable

Overview

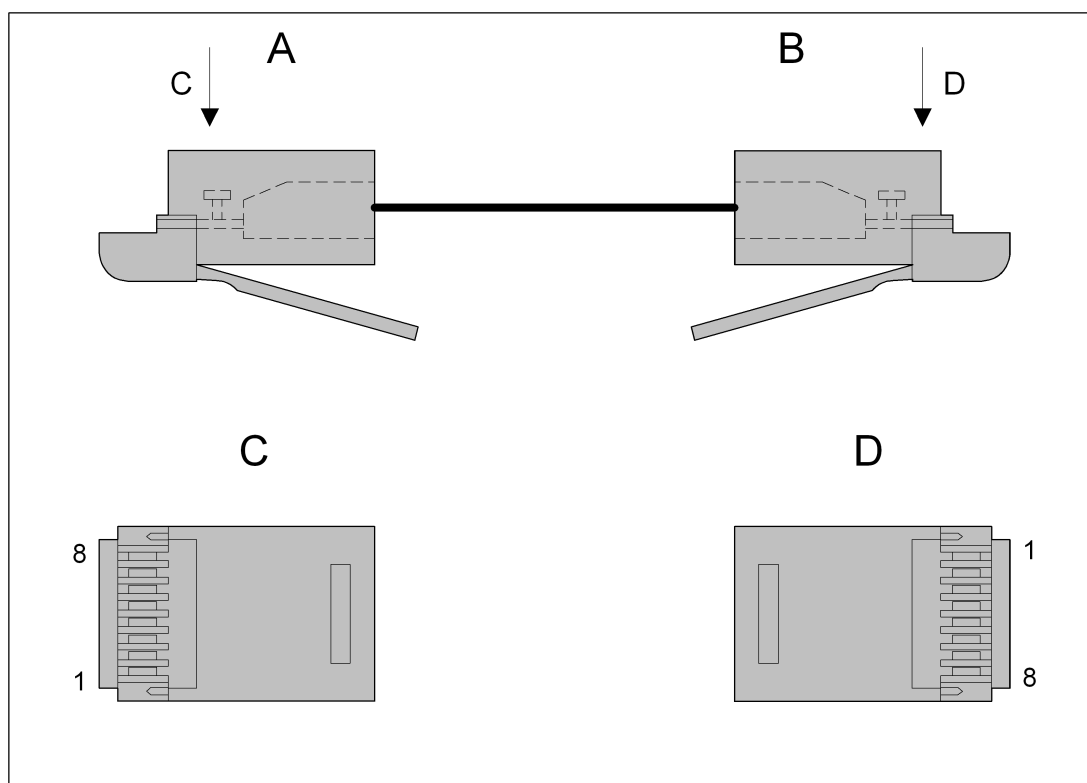
The control and switching card provides Ethernet maintenance interface on its front panel. It connects Ethernet switches or other network devices with network cables first, and then connects to the maintenance computers and provides out-of-band network management mode.

While laying out the network cables, select the network cables with proper length. Make sure that the cables are without bends or twists, and are properly bound together.

Network Cable Structure

Network cables are cat-5 cables with two ends of RJ45 interfaces, as shown in [Figure 5-16](#).

Figure 5-16 Network Cable Structure



Network Cable Connections

[Table 5-9](#) lists the relationship between network and core cables.

Table 5-9 Relationship Between Network Cables and Core Cables

End A	End B	Signal Definition
RJ45_1	RJ45_3 (Crossover)	TPOP
	RJ45_1 (Straight-through)	
RJ45_2	RJ45_6 (Crossover)	TPOP
	RJ45_2 (Straight-through)	
RJ45_3	RJ45_1 (Crossover)	TPIP
	RJ45_3 (Straight-through)	
RJ45_6	RJ45_2 (Crossover)	TPIN
	RJ45_6 (Straight-through)	

**Note:**

The network cable cannot be longer than 100 m.

5.7 Optical Fiber

Optical fibers can be used and led into the cabinet in the upper or lower cabling area or through the optical distribution frame. Each pigtail fiber should be tagged for identification of sending and receiving optical interfaces.

While connecting an optical fiber, make sure that the positioning pin on the connector is suitable for the positioning slot on the coupler. Make sure that there are no twists and bind the pigtail fibers appropriately. No other cables should be laid on the optical fibers. Protect optical fiber with protection tubes if necessary.

**Caution!**

Do not look at the connector while connecting the fiber.

This page intentionally left blank.

Chapter 6

Installation Check

Table of Contents

Overall Inspection	6-1
Power-ON/OFF Inspection	6-2

6.1 Overall Inspection

Overall inspection includes the following:

- Cabinet Inspection
- Cabling Inspection
- Shelf Power Supply Inspection
- Card Power Supply Inspection

Cabinet Inspection

Cabinet inspection includes the following:

- The vertical deviation of the cabinets should not exceed 3 mm.
- Make sure that the front, back, and side doors have been assembled.
- Make sure that all the screws have been tightened.
- Make sure that no parts of the cabinet fall off or are damaged. Connecting cables should be intact.
- Place the cabinet closely to the next one at the same level when they are arranged side by side.
- Place all the cabinets steadily and neatly with consistent gap between rows, both ends, and each point.
- Keep the cabinet clean and free of dust.

Cabling Inspection

Cabling inspection includes cable layout inspection, and power cable and grounding cable inspection.

- Cable layout inspection includes the following:
 - The turnings of cables must be even and smooth. Keep curves vertical or horizontal in a straight line.
 - Bind tail fibers inside the cabinet neatly and gently. Protect tail fibers between cabinets with sleeves.
 - Installation positions of cabling chutes and troughs must meet requirements of the construction drawings. Deviation from left to right cannot exceed 50 mm.

- Bind cables in toughs tightly and straightly. Fix cables at turnings or outlets with plastic clips.
- Power cable and grounding cable inspection includes the following:
 - The specifications of power cables and grounding cables must meet the design requirements.
 - The power cable and grounding cable must be one-piece cable without any joints.
 - Lay the power cable and grounding cable separately from other cables.
 - Connect the primary power to the power terminals on the cabinet. Make sure that the positive and negative polarities are connected properly.

Shelf Power Supply Inspection

Shelf power supply inspection includes the following:

- Check the shelf power cable voltage with a multimeter. Make sure that the shelf power cable voltage meets the power supply requirements.
- Power ON the power card to check the shelf power supply.
- Check the backplane power supply.

Card Power Supply Inspection

Card power supply inspection includes the following:

- Check whether the card is correctly inserted.
- Power ON the power card to check the card power supply.

6.2 Power-ON/OFF Inspection

This topic includes the following:

- Inspection Before Power-ON
- Power-ON Process
- Power-OFF Process

6.2.1 Inspection before Power-ON

Overview

After installing the cabinet successfully, conduct power-ON test. Check and confirm the internal and external status of the equipment before power-ON.

Temperature, Humidity, and Voltage Check

The temperature, humidity, and voltage check includes the following:

- Temperature: -5°C to 45°C
- Relative humidity: 10% – 90%
- DC voltage: -57 V DC to -41 V DC

Power and Internal Cables Check

The power and internal cables check includes the following:

- Check whether the connection of power and grounding cables are appropriate. Also check power polarities.
- Check whether the plug-in units are inserted in properly.
- Check whether jumpers of each card are connected properly.
- Check whether the cables are connected properly.
- Check whether the boot version of each card is correct.

Other Checks

Other checks include the following:

- Equipment labels must be correct, complete, and clear.
- The quantity of plug-in units is correct.
- All power switches are set to OFF.
- Fuses meet the specifications.
- Equipment are grounded properly and the grounding resistance meet technical specifications.
- Make sure that there is no short circuit.

6.2.2 Power-ON Process

Power-ON process includes the initial power-ON process and daily power-ON process.

Initial Power-ON Process

The initial power-ON process is as follows:

1. Ensure that all the switches are OFF.
2. Use a multimeter to measure resistance of the busbar to confirm whether any short-circuit exists between the power supply (-48 V) and the ground (GND, GNDP, -48 V GND). Make sure that GND, GNDP, and -48 V GND are connected.
3. In case of zero load, check the external power input end of the cabinet with the multimeter. Measure the voltage of the primary power supply. Confirm that it is within the permitted voltage range.
4. Turn ON the external power. Turn OFF the power immediately in case of any trouble. Power ON the device again after eliminating the trouble.
5. Turn ON the power distributor. Turn OFF the power immediately in case of any trouble. Power ON the device again after eliminating the trouble.
6. Plug in all the cards.
7. Turn ON the power. Turn OFF the power immediately in case of any trouble. Power ON the device again after eliminating the trouble.

Daily Power-ON Process

1. Turn ON the external power. Then, turn ON the power distributor. Turn OFF the power immediately in case of any trouble. Power ON the device again after eliminating the trouble.
2. Turn ON the secondary power of the power card. Turn OFF the power immediately in case of any trouble. Power ON the device again after eliminating the trouble.
3. Turn ON the power. Turn OFF the power immediately in case of any trouble. Power ON the device again after eliminating the trouble.

6.2.3 Power-OFF Process

The power-OFF process is as follows:

1. Turn OFF the secondary power of the power card.
2. Turn OFF the power distributor.
3. Turn OFF the external power.

Appendix A

Indoor Cabinet Grounding System Connection

Table of Contents

- Overview A-1
- Central Office Cabinet Grounding A-1

A.1 Overview

Grounding ensures secure running and protection for the system. Grounding provides the equipment with the capability of anti-electromagnetic interference, preventing the equipment from damage caused by lightning.

Before connecting the cabinet to the grounding network, make sure the following items are ready:

- The equipment room lightning protection devices are available.
- The equipment room grounding devices are installed.
- Tools for cable installation are in place.
- All the grounding cables and inter-cabinet cables are placed according to the layout of the indoor cabinet.

A.2 Central Office Cabinet Grounding

This topic includes the following:

- Central Office Cabinet Grounding
- Remote Office Cabinet Grounding

A.2.1 Central Office Cabinet Grounding

There are two types for connecting the central office cabinet to the grounding network:

- Type 1
- Type 2

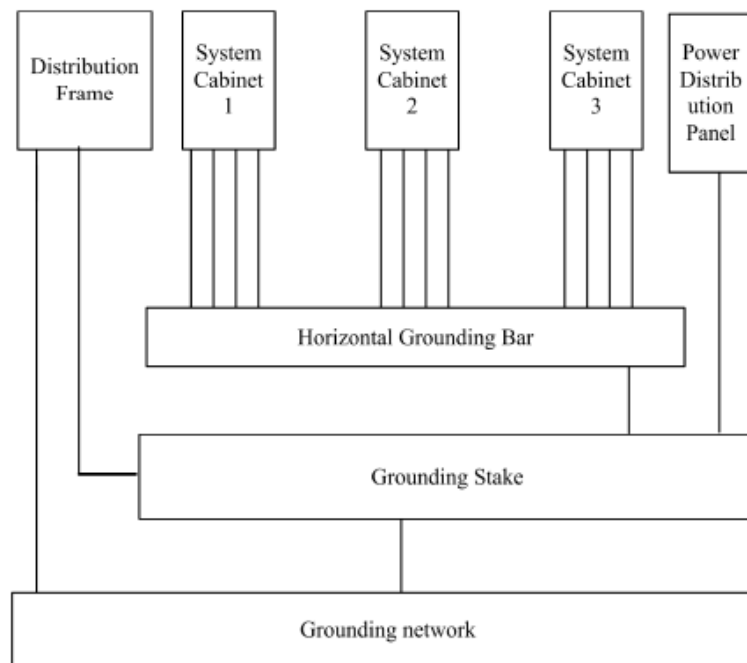
Type 1

- Cabinet grounding
 1. Connect the GNDP, -48 VGND, GND, and cabinet grounding cables from the busbar to the horizontal grounding bar.

2. Connect the horizontal grounding bar to the vertical grounding bar.
 3. Connect the vertical grounding bar to the grounding stake.
 4. Use the grounding inlead cable to connect the grounding stake to the grounding network.
- Distribution frame grounding
Connect the protection ground of the distribution frame to the grounding stake.
 - Power distribution cabinet grounding
Connect the power distribution cabinet to the grounding stake, which is connected to the grounding network through a grounding inlead cable.

Figure A-1 shows central office cabinet grounding (Type 1).

Figure A-1 Central Office Cabinet Grounding - Type 1



Type 2

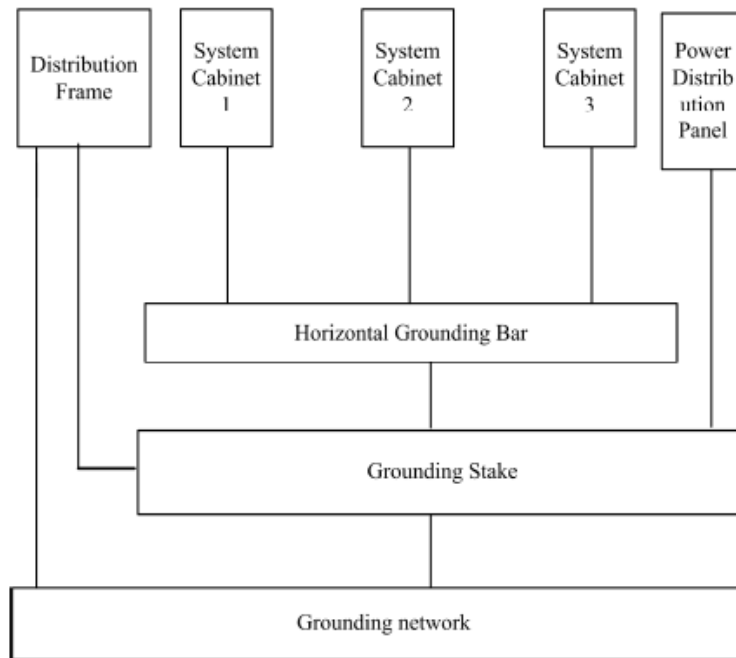
- Cabinet grounding
 1. Connect the GND, GNDP, -48 VGND, and cabinet from the busbar of the cabinet to the grounding copper bar of the cabinet.
 2. Use the multi-copper wire (grounding cable $\geq 95 \text{ mm}^2$) to connect the grounding copper bar to the horizontal grounding bar.
 3. Connect the horizontal grounding bar to the vertical grounding bar.
 4. Connect the vertical grounding bar to the grounding stake.
 5. Connect the grounding stake to the grounding network through the grounding inlead cable.
- Distribution frame grounding
Connect the protection ground of the distribution frame to the grounding stake.

- Power distribution cabinet grounding

Connect the power distribution cabinet to the grounding stake, which is connected to the grounding network through the grounding inlead cable.

Figure A-2 shows central office cabinet grounding (Type 2).

Figure A-2 Central Office Cabinet Grounding - Type 2



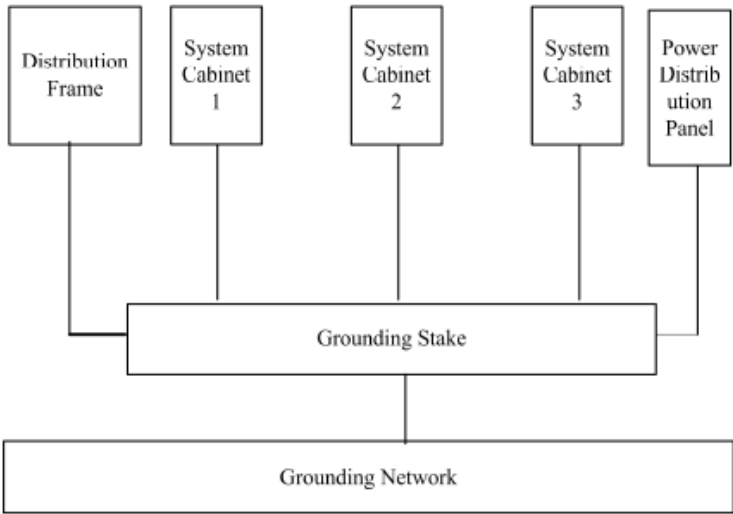
A.2.2 Remote Office Cabinet Grounding

The process of connecting the remote office cabinet to the grounding network is as follows:

1. Connect the GND, GNDP, -48 VGND, cabinet, and the power supply system to the grounding copper bar of the cabinet. Use multi-copper wire ($\geq 95 \text{ mm}^2$) to connect the grounding copper bar to the grounding stake.
2. Connect the distribution frame protection ground to the grounding stake.
3. If the power distribution panel is located on the grounding floor, directly connect the protection ground to the grounding network.
4. If the power distribution panel is located anywhere other than the ground floor, connect the corresponding protection ground to the grounding stake.
5. Connect the grounding stake to the grounding network through the grounding inlead cable.

Figure A-3 shows remote office cabinet grounding.

Figure A-3 Remote Office Cabinet Grounding



Appendix B

Grounding Network Requirements

Table of Contents

- Grounding Cable Requirements..... B-1
- Cabinet and Central Office..... B-2
- Outgoing/Incoming Cables from/to Central Office B-2

B.1 Grounding Cable Requirements

Grounding cable requirements are as follows:

- Material
- Cable length and cross-sectional area

Material

Use an insulated copper cable instead of the exposed wire as the grounding cable. Use the copper bar or galvanized flat steel as the busbar.

Cable Length and Cross-sectional Area

Table B-1 lists the length and cross-sectional area requirements for grounding cables.

Table B-1 Length and Cross-sectional Area Requirements

Cable	Requirements
Grounding bar	Copper bar ($\geq 40 \times 4 \text{ mm}^2$), galvanized flat steel ($\geq 50 \times 5 \text{ mm}^2$), or multi-copper wire (cross-sectional area $\geq 150 \text{ mm}^2$); solder the joint of the steel and copper.
	Use the grounding inlead to connect the grounding stake to the nearby grounding network.
Grounding inlead	Copper bar ($\geq 40 \times 4 \text{ mm}^2$), galvanized flat steel ($\geq 50 \times 5 \text{ mm}^2$), or multi-copper wire (cross-sectional area $\geq 120 \text{ mm}^2$); all materials must be insulated.
	The grounding inlead of the telecommunications office should be less than 30 meters.

Cable	Requirements
SPD for data cable and other types of SPD	The cross-sectional area of SPD for data cable is larger than 2.5 mm ² . The cable is multi-copper wire with the length less than 1 meter.
	The distance between the terminal of modular SPD for power supply and the phase line, and the zero line should be less than 0.5 meter.
	SPD grounding cable should be less than 1 meter and it should be grounded nearby.
Distribution frame protection cable	Multi-copper wire (≥ 95 mm ²)
Cabinet grounding cable	Multi-copper wire (≥ 35 mm ²)
GNDP	Multi-copper cable (≥ 16 mm ²), depending upon the maximum fault current.
-48 V GND and GND	Multi-copper cable (≥ 10 mm ²)
220 V protection grounding	Multi-copper cable (≥ 50 mm ²)

B.2 Cabinet and Central Office

The grounding requirements for cabinet and central office are as follows:

- The cabinet in the central office must be insulated.
- The central office must be at least 5 meters away from the lightning protection cable of the building.

B.3 Outgoing/Incoming Cables from/to Central Office

The grounding requirements for outgoing/incoming cables from/to central office are as follows:

- The power and telecommunication cables that go out of or come into the central office (station) must be covered with metal sleeves or must be laid in the metal tubes.
- Ground the metal sleeve or metal tube in the central office (station) or connect the cables to the grounding network. For fibers, ground the metal component in the fiber at the ground terminal.
- Lay the incoming cables underground at a depth of more than 0.7 meter.
- Connect all user cables to the SPD on distribution frame. Then, connect them to the telecommunication equipment.
- For incoming cables, it is not allowed to have idle pairs. Idle pairs of the incoming cables must be grounded. Idle pairs of cables for local telephony must be grounded on distribution frame.
- For incoming PCM cables, add SPD at the terminal. Idle pairs must be grounded. Ground the cover, common wire, and the sleeves of the power cables nearby.

- Lay the incoming low voltage power cables underground at a depth of more than 15 meters.
- Lay signal cables in the middle of the building. For vertical wiring, try to keep the cable at least 5 m away from the poles of the building (especially the poles at the four corners).
- Protect the unshielded cables with metal tubes. Keep the cables in the equipment room away from the lightning protection pole.
- Install the lightning protection SPD near the cable inlet of the equipment room. Connect the grounding cable of the lightning protection SPD separately from the grounding bar.
- Lay -48 V power cable, -48 V grounding cable, and protection grounding cable in parallel in the equipment room. Lay them with the signal cables (including user cables) in vertical direction.
- If the room for the access equipment is adjacent to the room for the mobile equipment, arrange the grounding cables in the access equipment room separately from the grounding cables of the mobile tower.
- Do not lay the AC power cable of the remote unit and remote modules in parallel near the signal cables (including user cables) and -48 V power cables.

This page intentionally left blank.

Appendix C

Lightning Proof Network

Table of Contents

- Lightning Requirements..... C-1
- Lightning Design..... C-1
- Lightning Protection Grounding in Central Office..... C-2
- Lightning Protection Grounding in Remote Office..... C-2

C.1 Lightning Requirements

Table C-1 lists the lightning protection levels of the power input and signal (Ethernet interface) input/output of the access network device.

Table C-1 Lightning Protection Level

Lightning Protection Port	Parameters
DC power	Cable – Ground: 1 kV DC power (1.2/50 μ s)
	Cable – Cable: 1 kV (1.2/50 μ s)
Ethernet interface	1 kV (1.2/50 μ s)

The lightning protection level of the equipment room should be Level B. The lightning protection level of the equipment should be Level C and Level D. Table C-2 lists the lightning protection device parameters at different levels.

Table C-2 Lightning Protection Device Parameters

Lightning Protection Level	Parameters	Lightning Protection Circuit
Level B	40 kA (8/20 μ s)	AC distribution box
Level C	20 kA (8/20 μ s)	DC distribution box
Level D	6000 V (combined wave)	-48 V

C.2 Lightning Design

The lightning protection devices of the equipment room must conform to the lightning protection standards. The cable inlets of the equipment room must be equipped with Level B lightning protection devices.

The lightning protection design for the cable inlets on the cabinet must conform to Level C and D standard. -48 V terminal is equipped with a 20 kA lightning protection device. The lightning protection design for outgoing user cables, E1 cables, and Ethernet cables must conform to Level D lightning protection standard.

C.3 Lightning Protection Grounding in Central Office

Take the following considerations when implementing the lightning protection grounding of the power systems in the Central Office:

- Lay the AC power cable underground at a depth of more than 15 meters.
- Install Level B lightning protection device with the AC distribution card. Connect the protection device grounding to the horizontal grounding bar.
- Install Level C lightning protection device in the DC distribution cabinet in the power supply system room. Connect the protection device grounding to the horizontal grounding bar.
- Install Level D lightning protection device in the rectifier. Connect the protection device grounding to the horizontal grounding bar through the DC distribution cabinet.

C.4 Lightning Protection Grounding in Remote Office

Take the following considerations when designing the lightning protection grounding of the power systems in a Remote Office:

- Lay the AC power cable under ground at a depth of more than 15 meters.
- Since both the AC distribution card and DC distribution cabinet are in same equipment room, the distance between Level B lightning protection device and Level C lightning protection device must meet the requirements of decoupling distance.
- If the protection grounding cable is arranged separately, the distance between Level B lightning protection device and Level C lightning protection device in the DC distribution cabinet should be more than 5 meters.
- If the protection grounding cable is arranged in parallel with the power cable, the distance between Level B lightning protection device and Level C lightning protection device in the DC distribution cabinet should be more than 15 meters.
- If the condition does not meet the requirements, install a decoupling inductance (calculate by $1.5 \mu\text{H/m}$) in front of Level C lightning protection device.
- Connect Level B lightning protection device installed on AC distribution card to the protection grounding bar in the equipment room.
- Connect Level C lightning protection device installed on DC distribution cabinet to the protection grounding bar in the equipment room.
- Connect Level D lightning protection device, installed on rectifier to the DC distribution card and then to the protection grounding bar in the equipment room.
- Connect -48 V grounding of the DC power distribution cabinet to working grounding bar. If there is no working grounding bar, connect it to the protection grounding bar.
- For the grounding cable of the AC distribution card equipped with Level B lightning protection device, use a multi-copper wire ($\geq 35 \text{ mm}^2$).

-
- For the protection grounding cable and working grounding cable of DC distribution cabinet equipped with Level C lightning protection device, use a multi-copper wire ($\geq 35 \text{ mm}^2$).
 - Connect the cables mentioned above to the protection grounding bar in the equipment room. Try to keep the length of the grounding cable as short as possible.

This page intentionally left blank.

Figures

Figure 1-1	Cabinet Appearance	1-2
Figure 1-2	Cabinet Appearance	1-2
Figure 1-3	Cabinet Configuration for 19-Inch Shelves	1-4
Figure 1-4	19-Inch Shelf Structure	1-5
Figure 1-5	19-Inch Shelf Structure	1-5
Figure 1-6	19-Inch Shelf Configuration.....	1-6
Figure 2-1	Installation Preparation Flow	2-2
Figure 2-2	Packing Box.....	2-8
Figure 2-3	Cabinet Inner Package	2-9
Figure 2-4	Shelf Packing Box.....	2-10
Figure 2-5	Card Packing Box	2-11
Figure 2-6	Opened Card Packing Box.....	2-12
Figure 3-1	Cabinet Installation Flowchart	3-2
Figure 3-2	Cabinet Base Structure.....	3-3
Figure 3-3	Base With Adjustable Height From 160 mm to 260 mm	3-3
Figure 3-4	Base With Adjustable Height From 260 mm to 460 mm	3-4
Figure 3-5	Base With Adjustable Height From 460 mm to 660 mm	3-4
Figure 3-6	Base Installation Flowchart	3-5
Figure 3-7	Drilling Diagram	3-6
Figure 3-8	Installing Base	3-7
Figure 3-9	Fixing Cabinet.....	3-8
Figure 3-10	Cabinet Installed on the Base	3-9
Figure 3-11	Foot Installation Flowchart.....	3-10
Figure 3-12	Drilling Diagram	3-11
Figure 3-13	Cabinet Installation Holes	3-13
Figure 3-14	Fixing Cabinet.....	3-14
Figure 3-15	Fixing Two Cabinets In Parallel	3-15
Figure 3-16	Fixing Multiple Cabinets In Parallel	3-16
Figure 3-17	Fixing Two Cabinets In Parallel	3-17
Figure 3-18	Fixing Multiple Cabinets In Parallel	3-18
Figure 3-19	Connecting Adjacent Cabinets	3-19
Figure 3-20	Installing Shelf	3-20

Figure 3-21	Installing Plug-in Box	3-21
Figure 4-1	Power and Grounding Cables Installation Flowchart	4-2
Figure 4-2	-48 V Power Cable Structure.....	4-3
Figure 4-3	Internal Power Cable Structure	4-3
Figure 4-4	Protection Grounding Cable Structure.....	4-4
Figure 4-5	19-Inch Power Distributor Dimensions	4-5
Figure 4-6	Power Distributor Internal Structure	4-6
Figure 4-7	Power Distributor Cable Connections.....	4-7
Figure 4-8	Internal Power Cable Layout.....	4-8
Figure 4-9	Cabinet Grounding Point.....	4-9
Figure 4-10	Protection Ground Connection	4-10
Figure 5-1	Uplink Cabling	5-2
Figure 5-2	Cables Bundling.....	5-4
Figure 5-3	Tie Cutting	5-4
Figure 5-4	Tie Space	5-5
Figure 5-5	Cable Tie Position.....	5-5
Figure 5-6	Cabling Technique In Cabinet	5-6
Figure 5-7	E1 Cable Structure	5-7
Figure 5-8	T1 Cable Structure.....	5-8
Figure 5-9	Monitoring Cable Routing.....	5-11
Figure 5-10	Entrance Control Monitoring Cable Structure	5-12
Figure 5-11	Smog Monitoring Cable Structure.....	5-13
Figure 5-12	Flood Monitoring Cable Structure.....	5-14
Figure 5-13	Temperature Monitoring Cable Structure	5-15
Figure 5-14	Humidity Monitoring Cable Structure	5-16
Figure 5-15	Serial Port Cable Structure	5-17
Figure 5-16	Network Cable Structure	5-18
Figure A-1	Central Office Cabinet Grounding - Type 1	A-2
Figure A-2	Central Office Cabinet Grounding - Type 2	A-3
Figure A-3	Remote Office Cabinet Grounding	A-4

Tables

Table 1-1	19-Inch Shelf Technical Specifications	1-6
Table 1-2	Card List.....	1-6
Table 1-3	Technical Standards	1-8
Table 1-4	Reliability and Security Specifications.....	1-11
Table 2-1	Operation Environment Requirements	2-4
Table 4-1	Internal Power Cable Pin Connections	4-3
Table 5-1	E1 Cable Pin Connections	5-7
Table 5-2	T1 Cable Pin Connections	5-9
Table 5-3	Entrance Control Monitoring Cable Pin Connections	5-12
Table 5-4	Smog Monitoring Cable Pin Connections.....	5-13
Table 5-5	Flood Monitoring Cable Pin Connections	5-14
Table 5-6	Temperature Monitoring Cable Pin Connections	5-15
Table 5-7	Humidity Monitoring Cable Pin Connections	5-16
Table 5-8	Serial Port Cable Connections.....	5-17
Table 5-9	Relationship Between Network Cables and Core Cables.....	5-19
Table B-1	Length and Cross-sectional Area Requirements	B-1
Table C-1	Lightning Protection Level.....	C-1
Table C-2	Lightning Protection Device Parameters	C-1

This page intentionally left blank.

Glossary

CES

- Circuit Emulation Service

DC

- Direct Current

DDF

- Digital Distribution Frame

EPON

- Ethernet Passive Optical Network

FE

- Fast Ethernet

GE

- Gigabit Ethernet

GPON

- Gigabit Passive Optical Network

IEC

- International Electrotechnical Commission

IP

- Internet Protocol

ISDN

- Integrated Services Digital Network

MAC

- Medium Access Control

MDF

- Main Distribution Frame

ODF

- Optical Distribution Frame

P2P

- Peer-to-Peer

PCM

- Pulse Code Modulation

PON

- Passive Optical Network

PSTN

- Public Switched Telephone Network

SDH

- Synchronous Digital Hierarchy

STM

- Synchronous Transport Module

TDM

- Time Division Multiplexing

Other ManualsLib Projects

 www.manualslib.com

 www.manualslib.de

 www.manualslib.es

 www.manualslib.fr

 www.manualslib.nl

 www.manualslib.mx

 www.manualslib.tech 30+ Languages